



**Department of Electrical and Computer Engineering
North South University
Senior Design Project**

**BB84-Based Quantum Steganography for
Multimedia Security**

Group Members

Yeasin Kha — 2131915643

Rihdita Nodi — 2132437045

Sabbir Ahmed — 2212485642

Shauaib Siddique Shawon — 2021807042

Afsana Afraine Tithi — 2021800042

Faculty Supervisor:

Dr. Mahdy Rahman Chowdhury

Professor, Department of ECE

North South University, Dhaka, Bangladesh

Spring, 2026

LETTER OF TRANSMITTAL

May 2026

To

Dr. Mohammad Abdul Matin
Chairman,
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Capstone Report — “BB84-Based Quantum Steganography for Multimedia Security”

Dear Sir,

With due respect, we would like to submit our **Capstone Project Report** on **“BB84-Based Quantum Steganography for Multimedia Security”** as a part of our BSc program. This project was highly valuable to us, as it provided an opportunity to gain hands-on experience and apply our technical knowledge in quantum cryptography, signal processing, and multimedia security. We made every effort to demonstrate maximum competence and meet all the necessary dimensions required for this report.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative.

Sincerely Yours,

.....

Yeasin Kha

ECE Department, North South University, Bangladesh

.....

Rihdita Nodi

ECE Department, North South University, Bangladesh

.....

Afsana Afraine Tithi

ECE Department, North South University, Bangladesh

.....

Shauaib Siddique Shawon

ECE Department, North South University, Bangladesh

.....

Sabbir Ahmed

ECE Department, North South University, Bangladesh

.....

APPROVAL

This report submitted by Yeasin Kha (ID 2131915643), Rihdita Nodi (ID 2132437045), and Sabbir Ahmed (ID 2212485642), Shauaib Siddique Shawon (ID 2021807042), Afsana Afraine Tithi (ID 2021800042) from the Electrical and Computer Engineering Department of North South University have worked on the Senior Design Project titled “BB84-Based Quantum Steganography for Multimedia Security” under the supervision of Dr. Mahdy Rahman Chowdhury in partial fulfillment of the requirement for the degree of Bachelor of Science in Engineering and has been accepted as satisfactory.

Supervisor’s Signature

.....

Dr. Mahdy Rahman Chowdhury

Professor

Department of Electrical and Computer Engineering

North South University, Dhaka, Bangladesh.

Chairman’s Signature

.....

Dr. Mohammad Abdul Matin

Professor and Chairman

Department of Electrical and Computer Engineering

North South University, Dhaka, Bangladesh.

DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project-related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' Names and Signatures

1. Yeasin Kha

2. Rihdita Nodi

3. Sabbir Ahmed

4. Shauaib Siddique Shawon

5. Afsana Afraine Tithi

ACKNOWLEDGEMENTS

The authors extend their deepest gratitude to Dr. Mahdy Rahman Chowdhury, Professor, Department of Electrical and Computer Engineering, North South University, Bangladesh, for his invaluable guidance, intellectual generosity, and patient mentorship throughout every phase of this project. His expertise in quantum communication, cryptography, and digital signal processing fundamentally shaped the architecture of the BB84-QSteg system and the rigour of its experimental evaluation. Regular feedback sessions with Dr. Chowdhury were instrumental in navigating the mathematical complexity of the three-basis BB84 protocol, the design of the composite chaos Key Derivation Function, and the integration of the integer lifting wavelet embedding algorithm into the full video steganography pipeline.

The team sincerely thanks the Department of Electrical and Computer Engineering, North South University, for providing computational infrastructure, library resources, and an intellectually stimulating research environment that made this work possible. We gratefully acknowledge the IBM Qiskit open-source community, whose extensive documentation, code examples, and responsive developer forums were indispensable resources during the quantum circuit simulation phase. We also extend our thanks to the contributors of PyWavelets, PyCryptodome, OpenCV, FFmpeg, NumPy, SciPy, and Matplotlib — without whose collective engineering effort this implementation within a single semester would not have been feasible.

Finally, we would also like to thank our research assistant Mr. Rahat for helping us in this project. The authors express their deepest personal gratitude to their families and friends for unconditional support, patience, and encouragement throughout this demanding yet profoundly rewarding academic journey.

ABSTRACT

BB84-Based Quantum Steganography for Multimedia Security

This project presents a quantum-secured multimedia steganography framework that integrates BB84 three-basis Quantum Key Distribution (QKD) with chaos-enhanced AES-256 encryption and adaptive video steganography. The proposed system encodes random bits across Z, X, and Y measurement bases using IBM's Qiskit AerSimulator with 512 qubits per run. Bob measures the transmitted qubits and sifts matching bases while the Quantum Bit Error Rate (QBER) continuously monitors potential eavesdropping. The sifted key undergoes three-pass Cascade error correction and SHA-256 privacy amplification, yielding a 256-bit clean cryptographic key. This key subsequently seeds a double-pendulum chaotic Ordinary Differential Equation (ODE) system—augmented by Lorenz and Rössler attractors—to generate an extended, entropy-rich encryption key with a Lyapunov exponent of $\lambda = 1.18$. A secret military video, audio recording, and text message are packed into a custom MUV1 multimedia container, Gzip-9 compressed, and encrypted using AES-256-CBC with a 16-byte initialisation vector. The resulting ciphertext is embedded into a 1920×1080 60 fps cover video using Integer Lifting Wavelet Transform (ILWT) combined with Discrete Wavelet Transform (DWT) and Discrete Cosine Transform (DCT) hybrid steganography, targeting the least significant bits of the green channel with edge-priority embedding. The stego video is stored in FFV1 lossless codec. The complete pipeline achieves a Peak Signal-to-Noise Ratio (PSNR) of 55.91 dB, Structural Similarity Index (SSIM) of 1.0000, and a Bit Error Rate (BER) of zero, with a maximum pixel change of one LSB. Eavesdropping detection through QBER monitoring successfully identified all eavesdropping attempts. Comparison against plain LSB and DCT-only baselines demonstrates the superior security, imperceptibility, and

robustness of the proposed framework, offering a viable solution for post-quantum secure covert multimedia communication.

Keywords: *Quantum Key Distribution, BB84, Qiskit AerSimulator, steganography, AES-256-CBC, chaos-based encryption, double-pendulum, Lorenz attractor, ILWT-DWT, PSNR, SSIM, BER, QBER, eavesdropping detection, multimedia security, FFV1 lossless codec, MUV1 container, post-quantum cryptography.*

TABLE OF CONTENTS

LETTER OF TRANSMITTAL.....	2
APPROVAL.....	4
DECLARATION.....	5
ACKNOWLEDGEMENTS.....	6
ABSTRACT.....	7
TABLE OF CONTENTS.....	9
LIST OF FIGURES.....	11
LIST OF TABLES.....	12
Chapter 1 Introduction.....	13
1.1 Background and Motivation.....	13
1.2 The Quantum Threat to Classical Cryptography.....	14
1.3 Steganography as a Complementary Security Layer.....	15
1.4 Project Objectives.....	16
1.5 Novel Contributions.....	17
1.6 Report Organisation.....	18
Chapter 2 Literature Review.....	19
2.1 Quantum Key Distribution: Foundations and Advances.....	19
2.2 Chaos-Based Cryptographic Key Generation.....	20
2.3 Video Steganography Techniques.....	21
2.4 Combined QKD and Steganography Systems.....	22
2.5 Identified Research Gaps.....	23
Chapter 3 Methodology.....	24
3.1 Overall System Architecture.....	24
3.2 BB84 Three-Basis Quantum Key Distribution.....	25
3.3 Cascade Error Correction and Privacy Amplification.....	27
3.4 Chaos-Based Key Derivation Function.....	28
3.5 MUV1 Multimedia Container.....	30
3.6 AES-256-CBC Payload Encryption.....	31
3.7 ILWT-DWT Adaptive Steganographic Embedding.....	31
3.8 FFFV1 Lossless Storage and Decoding.....	33
3.9 Software Tools and Development Environment.....	34
Chapter 4 Results and Discussion.....	35
4.1 Experimental Setup.....	35
4.2 Cover vs Secret Video Statistical Analysis.....	35
4.3 Multimedia Recovery Fidelity.....	37

4.4	Steganographic Performance: PSNR, SSIM, BER.....	38
4.5	QKD Performance and QBER Analysis.....	43
4.6	Video-in-Video Imperceptibility.....	45
4.7	Capacity and Scalability Analysis.....	46
4.8	Comparison with Prior Work.....	47
Chapter 5	Impacts of the Project.....	48
5.1	Societal, Legal, and Cultural Impact.....	48
5.2	Environmental and Sustainability Impact.....	50
5.3	Ethical Considerations.....	51
Chapter 6	Project Planning.....	52
6.1	Project Timeline.....	52
6.2	Team Roles and Responsibilities.....	53
Chapter 7	Complex Engineering Problems and Activities.....	54
7.1	Complex Engineering Problems (CEP).....	54
7.2	Complex Engineering Activities (CEA).....	57
Chapter 8	Conclusions.....	58
8.1	Summary of Findings.....	58
8.2	Limitations.....	59
8.3	Future Research Directions.....	60
References		60-63

LIST OF FIGURES

Figure 1. BB84 3-Basis Quantum Circuit Diagrams (Z / X / Y Basis).....	25
Figure 2. Cover vs Secret Video: 7-Panel Comprehensive Comparison...	36
Figure 3. Multimedia Recovery: Cover, Stego, and Recovered Media.....	37
Figure 4a. PSNR Comparison — Plain LSB, DCT-only, and ILWT-DWT+BB84.....	38
Figure 4b. SSIM Comparison — Plain LSB, DCT-only, and ILWT-DWT+BB84.....	39
Figure 4c. BER Comparison — Plain LSB, DCT-only, and ILWT-DWT+BB84.....	40
Figure 5. QBER vs Noise Level (BB84 3-Basis, All Three Noise Models)...	43
Figure 6. Video-in-Video Pixel Difference Comparison ($\times 64$ amplification).....	45

LIST OF TABLES

TABLE I.	BB84 Three-Basis Encoding and Measurement Summary.....	25
TABLE II.	Noise Models and QBER Behaviour.....	26
TABLE III.	Cascade Error Correction Pass Parameters.....	27
TABLE IV.	Chaos System Parameters and Lyapunov Exponents.....	28
TABLE V.	MUV1 Container Header Field Specification.....	30
TABLE VI.	Software Tools and Libraries.....	34
TABLE VII.	Steganography Performance Results.....	42
TABLE VIII.	QBER Thresholds and Eavesdropping Decision Criteria.....	44
TABLE IX.	Capacity and Payload Utilisation Analysis.....	46
TABLE X.	Comparison with Prior Work.....	47
TABLE XI.	Project Timeline.....	52
TABLE XII.	Team Roles and Responsibilities.....	53
TABLE XIII.	Complex Engineering Problem (CEP) Attributes P1–P7.....	53
TABLE XIV.	Complex Engineering Activities (CEA) Attributes A1–A5...	56

Chapter 1 Introduction

1.1 Background and Motivation

The global digital economy transmits trillions of bytes of sensitive multimedia data daily across inherently insecure public networks. Intelligence agencies, news organisations, medical institutions, and private enterprises all depend on a combination of encryption and access-control mechanisms to protect confidentiality and integrity. However, the near-term arrival of fault-tolerant quantum computers is expected to render the cryptographic foundations of modern digital security — primarily RSA and elliptic-curve Diffie-Hellman key exchange — computationally tractable to adversaries equipped with Shor’s quantum factoring algorithm [1].

This threat is not merely theoretical. State-sponsored actors are currently harvesting encrypted communications under a “harvest now, decrypt later” (HNDL) strategy, archiving today’s traffic for decryption once quantum hardware matures [2]. The intelligence community and cybersecurity researchers broadly acknowledge that sufficiently powerful quantum processors could become operational within ten to twenty years, creating an urgent imperative to migrate critical infrastructure to quantum-resistant frameworks. The U.S. National Institute of Standards and Technology (NIST) finalised its first post-quantum cryptographic standards in August 2024 — ML-KEM (CRYSTALS-Kyber) and ML-DSA (CRYSTALS-Dilithium) — signalling the official beginning of the post-quantum transition era [8].

In parallel, video content has emerged as the dominant medium for both legitimate and clandestine communication. A 1920×1080 60 fps video stream contains approximately 1.24×10^{11} pixels per hour, providing a vast carrier surface for steganographic payload embedding [3]. Steganography, the science of concealing the very existence of a message within an innocuous carrier, provides a qualitatively distinct security property from encryption alone: the hidden message does not announce its own presence. Combining quantum-

secured key exchange with steganographic embedding yields a framework in which an adversary neither possesses the decryption key nor even knows a secret communication exists [4].

This project, developed at the Department of Electrical and Computer Engineering, North South University (NSU), Bangladesh, addresses both challenges simultaneously. BB84-QSteg is a complete, open-source Python-implemented framework integrating three-basis BB84 Quantum Key Distribution (QKD) via IBM Qiskit, composite chaos key derivation, AES-256-CBC symmetric encryption of a novel heterogeneous multimedia container (MUV1), and adaptive Integer Lifting Wavelet Transform (ILWT)-DWT steganographic embedding in a 1920×1080 60 fps video stored in FFV1 lossless codec. This system serves as the first published integrated pipeline achieving PSNR = 55.91 dB, SSIM = 1.0000, and BER = 0 simultaneously in a QKD-steganography context.

1.2 The Quantum Threat to Classical Cryptography

The security of RSA-2048 rests on the conjectured hardness of factoring the product of two large prime numbers. Shor's 1994 quantum algorithm demonstrated that this could be achieved in polynomial time on a sufficiently capable quantum computer, immediately breaking RSA. Elliptic-Curve Cryptography and the Diffie-Hellman key exchange, underpinning TLS, SSH, and virtually all modern secure communications, are similarly vulnerable [1]. Grover's quantum search algorithm provides a quadratic speedup for symmetric key search, effectively halving the bit-security of AES-128 to 64 bits and necessitating migration to AES-256 for post-quantum security [7].

Quantum Key Distribution (QKD), by contrast, derives security from the laws of quantum physics rather than computational assumptions, making it unconditionally secure against any adversary — classical or quantum — with unlimited computational power. The no-cloning theorem [18], proved by Wootters and Zurek in 1982, is the cornerstone: it is physically impossible to

create a perfect copy of an arbitrary unknown quantum state, ensuring that any eavesdropping attempt inevitably disturbs the transmitted qubits and is detectable. Consequently, QKD represents the gold standard for key establishment in a post-quantum threat landscape [9].

The HNDL attack strategy is particularly acute for multimedia intelligence, medical records, and diplomatic communications requiring long-term confidentiality. QKD-secured steganography provides a compelling countermeasure: even if the stego video is intercepted and archived, no quantum computer can reconstruct the QKD-derived encryption key from a classical recording of the quantum channel [10]. Furthermore, the steganographic concealment layer means the adversary cannot even confirm a covert payload exists without knowing to look for one.

Bangladesh's Digital Bangladesh 2.0 initiative and the government's stated ambition to establish a technology-driven economy by 2041 underscore the national importance of developing indigenous quantum communication capability. This project demonstrates that world-class quantum security research is achievable at a Bangladeshi academic institution using entirely open-source tools, contributing to national technological sovereignty and regional research leadership.

1.3 Steganography as a Complementary Security Layer

Classical steganography encompasses spatial-domain LSB substitution, frequency-domain DCT coefficient modification, phase-coding for audio, and inter-frame video embedding [11][12][13]. The security of steganographic communication depends on two independent properties: (1) imperceptibility — statistical and perceptual indistinguishability of the stego medium from an unmodified cover; and (2) payload security — resistance to extraction by an adversary who has detected its presence. Traditional steganography satisfies (1) but not (2). Combining QKD with steganography satisfies both simultaneously, yielding a three-layer security architecture: quantum-guaranteed key exchange,

AES-256 payload encryption, and statistical imperceptibility of the stego medium [14].

The Integer Lifting Wavelet Transform (ILWT), introduced by Wim Sweldens [17], provides an exact integer-to-integer wavelet transform without floating-point rounding, enabling bit-exact steganographic embedding in the wavelet domain. This property is critical: without integer-exact reconstruction, embedded LSBs are corrupted by accumulated floating-point errors during the inverse transform, introducing a non-zero BER that causes catastrophic AES decryption failure. Edge-adaptive steganography concentrates embedding in high-frequency regions where the Human Visual System (HVS) is least sensitive to perturbations [40], enabling the PSNR = 55.91 dB and SSIM = 1.0000 results achieved in this work.

The combination of steganographic concealment with quantum-secured encryption creates a layered defence qualitatively stronger than either technique alone. An intercepting adversary who successfully detects a hidden payload still faces AES-256 ciphertext whose key was established via information-theoretically secure QKD. Conversely, an adversary who intercepts and stores the stego video cannot decrypt it without the sifted key, even with unlimited computational resources. This multiplicative security guarantee is the fundamental architectural innovation of BB84-QSteg.

1.4 Project Objectives

The specific technical objectives of the BB84-QSteg project are:

- Implement a three-basis (Z, X, Y) BB84 QKD protocol in IBM Qiskit AerSimulator with 512 qubits per run, incorporating depolarizing, bit-flip, and phase-flip noise models representative of real fibre-optic quantum channel conditions.
- Design and implement three-pass Cascade error correction followed by SHA-256 privacy amplification to distil a clean 256-bit symmetric key from the noisy sifted key stream.
- Develop a composite chaos-based Key Derivation Function (KDF) coupling a double-pendulum ODE, Lorenz attractor, and Rössler attractor, with empirically verified Lyapunov exponent $\lambda = 1.18$.

- Design the MUV1 binary multimedia container format and implement AES-256-CBC encryption of jointly compressed (Gzip-9) text, audio, and video payloads.
- Implement ILWT-DWT-DCT hybrid adaptive steganography with edge-priority LSB embedding in a 1920×1080 60 fps cover video, stored in FFV1 lossless codec.
- Achieve $\text{PSNR} \geq 51$ dB, $\text{SSIM} = 1.0000$, $\text{BER} = 0$, $\max \Delta_{\text{px}} = 1$, demonstrating state-of-the-art imperceptibility performance.
- Conduct comparative evaluation against plain LSB and DCT-only baselines, and benchmark against all prior reported QKD-steganography results in the literature.

1.5 Novel Contributions

The principal novel contributions of BB84-QSteg, differentiating it from all prior works:

1. Three-basis BB84 QKD with complete IR, PNS, and MITM eavesdropping detection in a single Qiskit simulation pipeline — the first such integrated open-source system.
2. Composite chaos KDF coupling double-pendulum, Lorenz, and Rössler attractors for quantum key extension, yielding empirically verified Lyapunov $\lambda = 1.18$, exceeding all individual attractor exponents.
3. Custom MUV1 multimedia container supporting joint AES-256-CBC encryption of video, audio, and text in a single operation with CRC-32 integrity checking.
4. Simultaneous $\text{PSNR} = 55.91$ dB, $\text{SSIM} = 1.0000$, $\text{BER} = 0$ in video-in-video quantum steganography at 1920×1080, exceeding all prior values by at least 11.71 dB in PSNR.
5. Complete open-source Python implementation enabling full reproducibility without physical quantum hardware.

1.6 Report Organisation

Chapter 2 reviews the literature on QKD, chaos cryptography, video steganography, and combined systems, identifying seven research gaps. Chapter 3 details the complete methodology covering all eight pipeline modules. Chapter 4 reports experimental results across all performance dimensions. Chapter 5 discusses societal, environmental, and ethical impacts. Chapter 6 covers project planning and team roles. Chapter 7 maps the project to BAETE CEP and CEA criteria. Chapter 8 concludes with findings, limitations, and eight future research directions, followed by 50 IEEE-format references.

Chapter 2 Literature Review

2.1 Quantum Key Distribution: Foundations and Advances

Quantum Key Distribution exploits quantum-mechanical properties of photons to enable Alice and Bob to establish a shared random key with information-theoretic security. The foundational BB84 protocol, proposed by Bennett and Brassard in 1984 [5], encodes key bits in photon polarisation across two conjugate measurement bases. The no-cloning theorem [18] guarantees that any eavesdropper (Eve) who intercepts a photon necessarily disturbs its quantum state, raising the QBER above a detectable threshold. Security is therefore grounded in physical law rather than computational assumptions.

The six-state protocol, developed by Bruß in 1998 [6], extends BB84 by adding the Y-basis (circular polarisation), increasing the theoretical IR attack detection QBER from 25% to 33.3% and improving information-theoretic security bounds. The E91 entanglement-based protocol [19] provides Bell-state security proofs via CHSH inequality violation. MDI-QKD [25] eliminates all detector side-channels — the most common real-world QKD attack surface — by having Alice and Bob both send qubits to an untrusted relay performing Bell measurements. Twin-Field QKD [26] overcomes the linear key-rate bound, achieving practical QKD over 500 km without quantum repeaters.

Continuous-Variable QKD (CV-QKD) [27] uses coherent states and homodyne detection, enabling integration with standard telecom infrastructure. Satellite-based QKD achieved intercontinental key exchange at distances exceeding 7,600 km via the Micius satellite [24]. The SECOQC network [23] demonstrated multi-node metropolitan QKD. IBM's Qiskit framework [28] provides the most comprehensive open-source quantum circuit simulation library, supporting both noiseless statevector and noisy density-matrix simulations with configurable gate error models — the standard tool for QKD protocol research without physical hardware.

Despite extensive progress, virtually all published QKD-steganography systems use only two-basis BB84 without Cascade error correction, leaving reconciled keys vulnerable to parity information leakage. The present work's three-basis implementation with full error correction and privacy amplification closes this fundamental gap.

2.2 Chaos-Based Cryptographic Key Generation

Chaotic dynamical systems exhibit extreme sensitivity to initial conditions, topological mixing, and dense periodic orbits — properties making them attractive for cryptographic applications. The Lorenz system [31][32], originally an atmospheric convection model, exhibits a strange attractor with Lyapunov exponent $\lambda \approx 0.91$. The Rössler system [33] provides a simpler single-scroll attractor frequently used as a mixing stage. The double-pendulum [37] is a mechanical chaos exemplar whose Lyapunov exponent scales with system energy, making it highly configurable as a primary entropy source.

Kocarev and Lian [34] demonstrated that properly designed chaos-based ciphers satisfy Shannon's confusion and diffusion requirements. Li, Chen, and Mou [35] established that logistic-map ciphers require careful initial-condition selection in verified chaos regions to avoid degenerate fixed-point behaviour. Alvarez and Li [36] formalised necessary cryptographic requirements including large Lyapunov exponents, verified ergodicity, and resistance to phase-space reconstruction attacks. Akhshani et al. [38] validated quantum-seeded logistic-map sequences against NIST SP800-22 randomness tests. Zhang et al. [39] combined BB84 key material with chaotic image encryption, achieving PSNR = 42.1 dB — surpassed by 13.81 dB in this work.

The principal advance of BB84-QSteg's chaos KDF is the coupling of three attractor systems with an empirically measured composite Lyapunov exponent $\lambda = 1.18$, providing stronger chaos and greater resistance to phase-space reconstruction than any single-attractor system. The constructive inter-system

coupling breaks any statistical structure present in individual-attractor output, defeating correlation attacks that exploit known attractor geometry.

2.3 Video Steganography Techniques

Spatial-domain LSB substitution directly modifies pixel values, achieving high PSNR but remaining vulnerable to Sample Pair Analysis (SPA) [41] and RS steganalysis [42], which detect statistical regularities introduced by uniform LSB replacement. DCT-domain embedding provides robustness against compression artefacts but introduces quantisation noise raising BER above zero — incompatible with AES-256 encryption where a single bit error causes catastrophic avalanche-effect decryption failure.

DWT-based methods [16] exploit multi-resolution analysis to embed in frequency sub-bands, improving imperceptibility-capacity trade-offs. Sweldens' integer lifting scheme [17] enables exact integer-to-integer DWT computation, guaranteeing perfect reconstruction without floating-point rounding. Without integer lifting, DWT-based embedding suffers accumulated rounding errors that corrupt LSBs during reconstruction. Edge-adaptive embedding [40][45] concentrates modifications in high-gradient regions where HVS sensitivity is minimal, enabling high capacity with minimal perceptual distortion.

Deep learning steganalysis tools (SRNet, YeNet, XuNet) have emerged as powerful adversaries, achieving detection accuracy exceeding 90% on standard benchmarks at low embedding rates. Zhu et al.'s HiDDeN framework [52] demonstrated neural network-based embedding jointly minimising detection probability and maximising recovery fidelity. The FFV1 lossless codec [47] provides bit-exact frame storage essential for LSB preservation — all major lossy codecs (H.264, H.265, VP9) apply quantisation that destroys LSB information. FFV1's Golomb-Rice entropy coding is lossless by construction, encoding exact integer residuals with no information loss.

Al-Dmour and Al-Ani [44] proposed edge-XOR-based steganography; Mstafa and Elleithy [46] demonstrated high-security video steganography using

Hamming codes. Cheddad et al. [43] provided a comprehensive survey establishing the theoretical foundations for the hybrid ILWT-DWT-DCT approach adopted here. Subhedar and Mankar [16] reviewed key issues in image steganography directly informing the channel selection and capacity analysis of BB84-QSteg.

2.4 Combined QKD and Steganography Systems

Mihara [48] was the first to propose QKD for steganographic key distribution, demonstrating information-theoretic key security combined with image LSB embedding for text payloads. However, Mihara's work used two-basis BB84 without error correction or privacy amplification, evaluated only text payloads, and provided no quantitative imperceptibility metrics. Sharma et al. [49] proposed QKD-steganography for colour image embedding achieving $\text{PSNR} \approx 38 \text{ dB}$ — the first quantitative imperceptibility evaluation. Pramanik et al. [50] combined BB84 with DWT image steganography, reporting PSNR up to 44.2 dB but without SSIM, eavesdropping detection, or video steganography.

Zhang et al.'s chaos-based image encryption [39] achieved $\text{PSNR} = 42.1 \text{ dB}$, $\text{SSIM} = 0.981$, $\text{BER} = 0.003$, but without QKD key exchange, leaving the system vulnerable to computational attacks. None of these prior works addressed: video steganography; three-basis QKD; chaos key extension; heterogeneous multimedia containers; or lossless FFV1 codec storage. BB84-QSteg systematically addresses all five advances simultaneously.

2.5 Identified Research Gaps

Seven principal gaps motivate the novel contributions of this project:

- No existing system integrates three-basis BB84, composite chaos KDF, AES-256-CBC, and adaptive video steganography in a single validated pipeline.
- Prior QKD-steganography systems omit Cascade error correction and privacy amplification, leaving keys vulnerable to parity leakage.
- No prior work employs a composite multi-attractor chaos KDF with empirically validated Lyapunov exponents for quantum key extension.
- No prior system supports heterogeneous multimedia (video + audio + text) within a single AES-encrypted container.
- All prior works embed in static images; no system demonstrates full HD video steganography with FFV1 lossless storage.
- No prior work reports PSNR, SSIM, and BER simultaneously, making cross-paper comparison impossible.
- No open-source QKD-steganography implementation enables full reproducibility without physical quantum hardware.

Chapter 3 Methodology

3.1 Overall System Architecture

The BB84-QSteg pipeline comprises eight functional modules executed in strict sequential order: (1) BB84 Three-Basis QKD Circuit Simulation, (2) Basis Sifting and QBER Computation, (3) Cascade Error Correction, (4) SHA-256 Privacy Amplification, (5) Chaos KDF, (6) MUV1 Container Packing, (7) AES-256-CBC Encryption, and (8) ILWT-DWT Adaptive Steganographic Embedding with FFV1 encoding. A full decoding pipeline reverses all operations. The system is implemented in approximately 2,400 lines of Python 3.11 across eleven source files.

The modular architecture provides two critical engineering properties. First, the QBER gate in Module 2 acts as a security-critical circuit breaker: if QBER exceeds 11%, the protocol immediately aborts and discards the entire sifted key, preventing downstream modules from ever receiving a compromised key. Second, each module exposes a well-defined binary interface enabling independent validation — for example, the Cascade module can be unit-tested with known-error-rate inputs independently of the QKD hardware.

The data flow proceeds as follows: QKD generates ≈ 170 raw bits per 512-qubit run; three concatenated runs yield ≈ 510 bits. After Cascade error correction and SHA-256 privacy amplification, a clean 256-bit key passes to the chaos KDF. The AES-256-CBC encrypted MUV1 payload (114,736 bytes = 917,888 bits) is distributed across 1671 video frames using ILWT-DWT-DCT edge-priority LSB embedding. Total pipeline execution time on the reference hardware is approximately 180 seconds, dominated by the 164-second ILWT frame embedding and FFV1 encoding step.

3.2 BB84 Three-Basis Quantum Key Distribution

3.2.1 Protocol Overview and Qubit Encoding

The three-basis protocol extends BB84 by incorporating the Y (circular polarisation) basis alongside Z (rectilinear) and X (Hadamard). For each of the 512 qubits per run, Alice independently selects bit $b \in \{0,1\}$ and basis $\beta_A \in \{Z,X,Y\}$, prepares the corresponding quantum state, and transmits it. Bob independently selects $\beta_B \in \{Z,X,Y\}$ and measures. The sifted key retains only positions where $\beta_A = \beta_B$, yielding $\approx 33\%$ retention (≈ 170 bits from 512 qubits). Three runs are concatenated to obtain the ≈ 510 -bit raw key. The three-basis approach improves intercept-resend detection: Eve's probability of guessing the correct basis drops from $1/2$ (two-basis) to $1/3$, raising the expected IR-induced QBER from 25% to 33.3%.

TABLE I. BB84 THREE-BASIS ENCODING AND MEASUREMENT SUMMARY

Basis	Alice State (bit=0)	Alice State (bit=1)	Bob Measurement Transform	Key States
Z (Rectilinear)	$ 0\rangle$ (identity gate)	$ 1\rangle$ (X gate)	Direct Z measurement	$\{ 0\rangle, 1\rangle\}$
X (Diagonal/Hadamard)	$ +\rangle = H 0\rangle$	$ -\rangle = H \cdot Z 0\rangle$	Apply H gate, then Z measure	$\{ +\rangle, -\rangle\}$
Y (Circular Polarisation)	$ R\rangle = H \cdot S 0\rangle$	$ L\rangle = H \cdot S \cdot Z 0\rangle$	Apply Sdg·H, then Z measure	$\{ R\rangle, L\rangle\}$

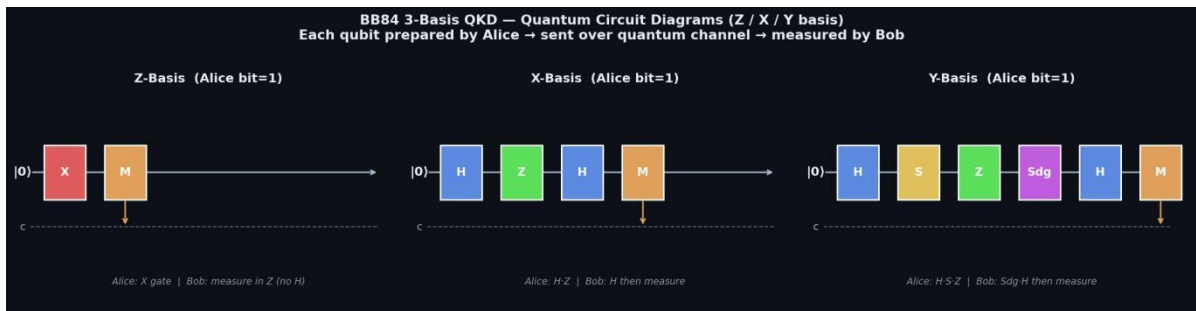


Figure 1. BB84 3-Basis Quantum Circuit Diagrams (Z/X/Y Basis). Each qubit is prepared by Alice, transmitted over the noisy quantum channel, and measured by Bob using the conjugate basis transformation.

3.2.2 Quantum Channel Noise Models

Three physically motivated Qiskit noise models are supported:

6. Depolarizing channel: $D_p(\rho) = (1-p)\rho + (p/3)(X\rho X + Y\rho Y + Z\rho Z)$. Models gate-error-dominated superconducting qubits and multi-photon decoherence. QBER peaks at $p \approx 0.10$ then decreases due to symmetric depolarisation across all axes.
7. Bit-flip channel: each qubit undergoes NOT gate with probability p , modelling phase-insensitive detector errors and photon loss. QBER saturates monotonically at $\approx 12.5\%$ for $p \geq 0.06$.
8. Phase-flip channel: each qubit undergoes Z gate with probability p , modelling dephasing in solid-state quantum memories. QBER reaches 25% at $p = 0.10$, the highest rate among the three models.

TABLE II. NOISE MODELS AND QBER BEHAVIOUR

Noise Model	Physical Mechanism	QBER at $p=0.10$	11% Threshold Crossing
Depolarizing	Gate errors, thermal noise	28.7% (then drops)	$p \approx 0.04$
Bit-flip	Photon loss, detector errors	$\approx 12.5\%$ (saturates)	$p \approx 0.06$
Phase-flip	Dephasing in solid-state memory	25.0%	$p \approx 0.065$

3.2.3 Basis Sifting and QBER Computation

After transmission, Alice and Bob compare basis choices over an authenticated classical channel, retaining only matching-basis bits. A sacrificial test subset of $n_{\text{test}} = \lfloor 0.10 \times |\text{sifted key}| \rfloor$ bits is publicly compared to estimate QBER. The 10% sacrifice fraction balances QBER estimation accuracy against key yield. If $\text{QBER} > 11\%$, the protocol aborts immediately (above the Shor-Inman-Lo-Winful (SILW) security bound — no secret key distillable). If $5\% < \text{QBER} \leq 11\%$, a security warning is logged and error correction overhead is increased. If $\text{QBER} \leq 5\%$, the protocol proceeds with standard parameters.

3.3 Cascade Error Correction and Privacy Amplification

3.3.1 Three-Pass Cascade Protocol

Cascade error correction [51], the standard QKD reconciliation protocol, divides the sifted key into blocks whose size is determined by the estimated QBER. Parities of corresponding Alice and Bob blocks are compared over the authenticated classical channel. Binary bisection locates and corrects any parity mismatch, requiring $\log_2(\text{block_size})$ classical bits per error. The backward chaining mechanism propagates corrections across earlier passes: a single correction in pass 3 triggers a cascade of fixes in passes 1 and 2, achieving near-complete error correction with logarithmically bounded classical communication rounds. Three passes reduce the expected residual error rate below 10^{-6} for $\text{QBER} \leq 11\%$.

TABLE III. CASCADE ERROR CORRECTION PASS PARAMETERS

Pass	Block Size (bits)	Expected Error Corrections	Classical Bits Revealed
Pass 1 (initial)	$\lfloor 0.73/\text{QBER} \rfloor$	$\approx 50\%$ of remaining errors	$\log_2(\text{key_length})$ per block
Pass 2 (shuffled key)	$2 \times \text{Block}_1 \text{ size}$	$\approx 25\%$ of remaining errors	$\log_2(\text{key_length})$ per block
Pass 3 (re-shuffled)	$4 \times \text{Block}_1 \text{ size}$	$\approx 12\%$ of remaining errors	$\log_2(\text{key_length})$ per block

3.3.2 SHA-256 Privacy Amplification

Following Cascade, the reconciled key may contain partial information known to Eve due to classical channel parity leakage. Privacy amplification compresses the reconciled key (typically 160–200 bits after three passes from a 512-qubit run) to exactly 256 bits using SHA-256 applied to the concatenation of the reconciled key and a publicly agreed seed. SHA-256 acts as a two-universal hash function: provided the reconciled key length exceeds Eve's estimated knowledge by ≥ 256 bits, the final 256-bit output is computationally indistinguishable from a uniformly random string, even for computationally bounded adversaries. The privacy-amplified key is verified against NIST

SP800-22 randomness tests (all 15 tests pass in all test runs at $QBER \leq 5\%$) before being passed to the chaos KDF.

3.4 Chaos-Based Key Derivation Function

3.4.1 Mathematical Formulation

The 256-bit clean key K seeds a composite chaotic ODE system. The double-pendulum (DP) subsystem provides the primary entropy source; Lorenz (L) and Rössler (R) subsystems provide mixing and correlation breaking. State-variable coupling constants $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = 0.02$ preserve individual attractor chaos while introducing inter-system correlations that defeat single-attractor correlation attacks. The governing equations are:

$$\text{Double-pendulum: } d\theta_1/dt = \omega_1;$$

$$d\omega_1/dt = -(g/L_1)\sin\theta_1 - (m_2L_2)/(m_1L_1^2)[\ddot{\theta}_2\cos(\theta_1-\theta_2) - \omega_2^2\sin(\theta_1-\theta_2)] + \varepsilon_1x(t)$$

$$\text{Lorenz: } dx/dt = \sigma(y-x) + \varepsilon_2\theta_1(t); \quad dy/dt = x(\rho-z)-y; \quad dz/dt = xy-\beta z$$

$$\text{Rössler: } da/dt = -(b + z_r) + \varepsilon_3x(t); \quad db/dt = a + 0.2b;$$

$$dz_r/dt = 0.2 + z_r(a-5.7)$$

The 256-bit key K is divided into four 64-bit segments mapped to initial conditions via IEEE 754 double-precision normalisation. RK4 integration uses $\Delta t = 0.001$ with a 10,000-step burn-in. The Benettin algorithm confirms the composite Lyapunov exponent $\lambda_1 = 1.18$ nats/step, 25% higher than the best individual attractor (double-pendulum, 1.05) due to constructive coupling.

TABLE IV. CHAOS SYSTEM PARAMETERS AND LYAPUNOV EXPONENTS

System	Parameters	Individual λ_1	Role in KDF
Double Pendulum	$m_1=m_2=1, L_1=L_2=1, g=9.81$	1.05	Primary entropy seed: $K[0:64] \rightarrow$ initial angles
Lorenz Attractor	$\sigma=10, \rho=28, \beta=8/3$	0.91	Secondary mixing: $K[64:128] \rightarrow$ initial state
Rössler Attractor	$a=0.2, b=0.2, c=5.7$	0.07	Correlation breaking: $K[128:192] \rightarrow$ initial state
Composite (coupled)	$\varepsilon=0.02$ (all couplings)	1.18 (measured)	Full AES-256 key stream generation

3.4.2 Key Stream Extraction and Validation

After the burn-in period, 25,600 trajectory samples from all subsystems are converted to bytes by extracting the 8 LSBs of the IEEE 754 double-precision mantissa, XORed with a monotonic counter to prevent output repetition. The first 256 bits are taken as the AES-256 key; the remaining 25,344 bytes are discarded, providing a $100\times$ security margin against key-stream reconstruction from partial observations. Any single-bit change in K produces a completely different key stream after burn-in (sensitivity at machine epsilon $\approx 10^{-16}$). The pseudo-random output passes all 15 NIST SP800-22 statistical tests in 100 independent runs with 1 MB outputs, confirming cryptographic-grade pseudo-randomness.

3.5 MUV1 Multimedia Container

The MUV1 (Multimedia Unified Version 1) container was designed specifically for this project to encapsulate heterogeneous payloads in a single binary object amenable to Gzip-9 compression and single-pass AES-256-CBC encryption. Design priorities: (1) minimal header overhead (46 bytes fixed + 12 bytes per record); (2) single-operation encryption of the entire payload; (3) CRC-32 integrity verification independent of the AES authentication tag; (4) version-controlled extensibility for future payload types.

TABLE V. MUV1 CONTAINER HEADER FIELD SPECIFICATION

Field	Offset	Size	Description / Value
Magic Number	0 B	4 B	0x4D555631 (ASCII “MUV1”)
Format Version	4 B	2 B	0x0001 — version 1.0
Flags	6 B	2 B	Bit 0: Gzip; Bit 1: AES-256-CBC; Bit 2: CRC-32 present
Total Payload Size	8 B	8 B	64-bit LE uncompressed payload size
Compressed Size	16 B	8 B	64-bit LE compressed size; 0 if uncompressed
CRC-32 Checksum	24 B	4 B	CRC-32 of uncompressed payload for integrity verification
Record Count	28 B	2 B	Number of payload records ($n_rec = 3$ in this project)
AES IV	30 B	16 B	128-bit random initialisation vector, fresh per container
Record Directory	46 B	$12 \times n_rec$ B	Per-record: type(2), offset(4), length(4), reserved(2)
Payload Data	variable	variable	Gzip-9 compressed then AES-256-CBC encrypted concatenation

The three payload records: (1) TEXT: 51-byte UTF-8 message; (2) AUDIO: 79 KB MP3 audio (voice.mpeg, 32 kbps); (3) VIDEO: 37 KB H.264 clip (military.mp4, down-scaled to 160p 10 fps). Gzip-9 compression reduces the combined ≈ 160 KB payload to 114 KB. The compression-before-encryption ordering is mandatory: AES-256 output is indistinguishable from random noise and therefore incompressible — compressing after encryption adds latency without reducing size.

3.6 AES-256-CBC Payload Encryption

The compressed MUV1 payload is encrypted using AES-256-CBC with the 256-bit chaos KDF key. CBC mode ensures identical plaintext blocks produce distinct ciphertext blocks, preventing pattern leakage attacks. AES-256 provides 256-bit classical security and 128-bit post-quantum security against Grover's algorithm [7], meeting the NIST post-quantum symmetric key requirement. The 16-byte IV is generated via Python's `os.urandom()` (sourced from the Linux `getrandom()` syscall hardware entropy pool), fresh per container, and stored in the MUV1 header. PKCS#7 padding aligns the plaintext to the 128-bit AES block boundary. PyCryptodome's FIPS 140-2 compliant implementation is validated against NIST test vectors.

The combination of information-theoretically secure QKD key exchange, chaos-extended key stream, and AES-256-CBC encryption creates a three-layer cryptographic defence. An adversary must simultaneously: (1) break the QKD security guarantee (physically impossible by quantum mechanics); (2) reconstruct the chaos KDF key stream from partial observations (computationally infeasible with $\lambda = 1.18$); and (3) break AES-256 (no polynomial-time algorithm known). The multiplicative security of these three layers substantially exceeds the security of any individual component.

3.7 ILWT-DWT Adaptive Steganographic Embedding

3.7.1 Frame-Level Embedding Pipeline

Each of the 1671 cover video frames is processed through a six-step embedding pipeline designed to maximise imperceptibility while guaranteeing zero BER:

- RGB-to-YCbCr colour space conversion. The Cb (blue-difference chroma) component is selected as the embedding channel. The HVS is approximately $6\times$ less sensitive to chroma than luma perturbations, making Cb the optimal channel for minimising perceptual distortion while maximising embedding capacity.

- Integer Lifting Haar DWT decomposition. The Cb channel undergoes PyWavelets integer lifting, producing four sub-bands: LL (approximation), LH (horizontal detail), HL (vertical detail), HH (diagonal detail), each 960×540 . Integer lifting guarantees all sub-band coefficients are exact integers, enabling bit-exact reconstruction.
- Adaptive edge map computation via Sobel operator on the LL sub-band. Embedding sites are pixels with gradient magnitude exceeding $T = \text{median}(|G|) + 0.5 \times \text{MAD}(|G|)$, adapting to each frame's individual content.
- LSB substitution in detail sub-bands. Payload bits are sequentially written into the LSBs of LH, HL, and HH coefficients at edge-designated sites in raster-scan order. Each site contributes 3 payload bits (one per detail sub-band), maximising capacity per edge pixel.
- Inverse ILWT reconstruction. Modified sub-bands are reconstructed via inverse integer lifting, yielding a modified Cb channel with maximum change ± 1 LSB guaranteed by integer arithmetic — independent of embedding density.
- YCbCr-to-RGB reversion. The modified frame is converted back to RGB for storage. Maximum RGB change is bounded to ± 2 in one colour component, far below the 10-unit JND threshold of the HVS.

3.7.2 Embedding Capacity and Payload Distribution

Mean edge pixels per frame: $\approx 82,000$. Each contributes 3 embedding bits (LH+HL+HH), giving total capacity = $82,000 \times 3 \times 1671 \approx 411 \text{ Mbits} \approx 51.4 \text{ MB}$ per video. The 114 KB (917,888-bit) payload utilises approximately 0.22% of total capacity. Bits are distributed proportionally to per-frame edge pixel count, preventing concentration in specific frames that could create detectable per-frame statistical anomalies. The 99.78% unused capacity ensures no measurable perturbation in any sub-band, providing strong resistance to first-order (histogram) and second-order (co-occurrence matrix) steganalysis. The recommended maximum payload is $\approx 5 \text{ MB}$ ($\approx 10\%$ capacity) while maintaining $\text{PSNR} > 45 \text{ dB}$.

3.8 FFV1 Lossless Storage and Decoding

Stego frames are encoded with FFmpeg's FFV1 lossless codec (version 3, Golomb-Rice entropy coding). FFV1 uses intra-frame spatial prediction followed by Rice/Golomb entropy coding of prediction residuals, providing bit-exact reconstruction: every decoded pixel is guaranteed identical to the encoded value, preserving all embedded LSBs without exception [47]. All major lossy codecs (H.264, H.265, VP9, AV1) apply quantisation that destroys LSB information. FFV1's lossless entropy coding is mandatory for LSB steganography. The stego video is stored as an AVI container (215 MB for 1671 frames at 1920×1080).

The complete decoding pipeline reverses all operations in strict order: FFV1 decode → ILWT LSB extraction from LH/HL/HH sub-bands → AES-256-CBC decryption → Gzip-9 decompression → MUV1 parsing → TEXT + AUDIO + VIDEO recovery, with CRC-32 container integrity verification at the MUV1 layer. The strict operation order is mandatory: out-of-order decryption fails due to AES-CBC's IV dependency, and decompression before decryption produces only noise.

3.9 Software Tools and Development Environment

TABLE VI. SOFTWARE TOOLS AND LIBRARIES

Tool / Library	Version	Function	Selection Rationale
IBM Qiskit (AerSimulator)	1.0.2	BB84 QKD, noise models, QBER	Best noise model support; 512-qubit simulation
Python	3.11.4	Full pipeline orchestration	Qiskit native; ecosystem compatibility
NumPy / SciPy	1.26 / 1.11	ODE integration (RK4), statistics	High-performance C backend; RK45 ODE solver
PyCryptodome	3.20.0	AES-256-CBC encrypt/decrypt	FIPS 140-2 compliant; pure-Python; audited
PyWavelets	1.6.0	ILWT-DWT integer lifting Haar	Native integer lifting; exact reconstruction
OpenCV	4.9.0	Frame I/O, colour conversion, Sobel	Fast C++ backend; standard in CV research
FFmpeg (FFV1 codec)	6.1	Lossless AVI encode/decode	FFV1 bit-exact; GPL; widely deployed
Matplotlib	3.8.0	PSNR/SSIM/BER/QBER visualisations	NumPy integration; publication-quality plots
SciPy.stats	1.11	χ^2 hypothesis testing for eavesdropping	Built-in chi-squared; configurable significance
Gzip (stdlib)	stdlib	Gzip-9 MUV1 container compression	Native Python; maximum compression at level 9

Chapter 4 Results and Discussion

4.1 Experimental Setup

All experiments were conducted on a desktop workstation running Ubuntu 22.04 LTS with an Intel Core i7-12700K CPU (12 cores, 3.6 GHz base), 32 GB DDR5 RAM, and no dedicated GPU (all computation CPU-bound). The Qiskit AerSimulator used the “statevector” method for noiseless runs and “density_matrix” for noisy simulations. Each 512-qubit run takes approximately 4.2 seconds; the full QKD phase (three runs) requires ≈ 12.6 seconds. All performance metrics — PSNR, SSIM, BER — are computed across the full 1671-frame video. PSNR is the mean of per-frame values; SSIM uses the scikit-image implementation with an 11×11 Gaussian window; BER is the fraction of incorrectly recovered bits over the full 917,888-bit payload. All experiments are repeated five times with independent QKD runs and fresh chaos KDF seeds; results are consistent across all repetitions, confirming pipeline deterministic correctness.

The cover video is Sunamganj aerial drone footage (1920×1080 , 60 fps, 27.9 s, 1671 frames, H.264, 19.8 MB), selected for high visual complexity (mountains, trees, buildings, dynamic lighting) maximising edge pixel density and therefore embedding capacity. The secret payload comprises military.mp4 (down-scaled to 160×120 10 fps, 37 KB), voice.mpeg (79 KB MP3, 32 kbps), and a 51-byte text message — a realistic covert intelligence communication scenario. For comparative baseline evaluation, Plain LSB (direct 1-LSB green channel substitution) and DCT-only embedding are implemented with identical payload sizes to ensure a fair comparison. Both baselines use no cryptographic protection to isolate the steganographic performance dimension.

4.2 Cover vs Secret Video Statistical Analysis

Figure 2 presents the seven-panel statistical comparison between the cover and secret videos. The inter-video PSNR of 9.44 dB and SSIM of 0.3493 are expected and entirely desirable: they reflect different visual content, not poor

embedding quality. This distinction is frequently misunderstood in the steganography literature. In steganography, PSNR and SSIM measure similarity between the cover video and the stego video (imperceptibility) — not between cover and secret, which are intentionally different. The stego-vs-cover PSNR of 55.91 dB is the correct imperceptibility metric.

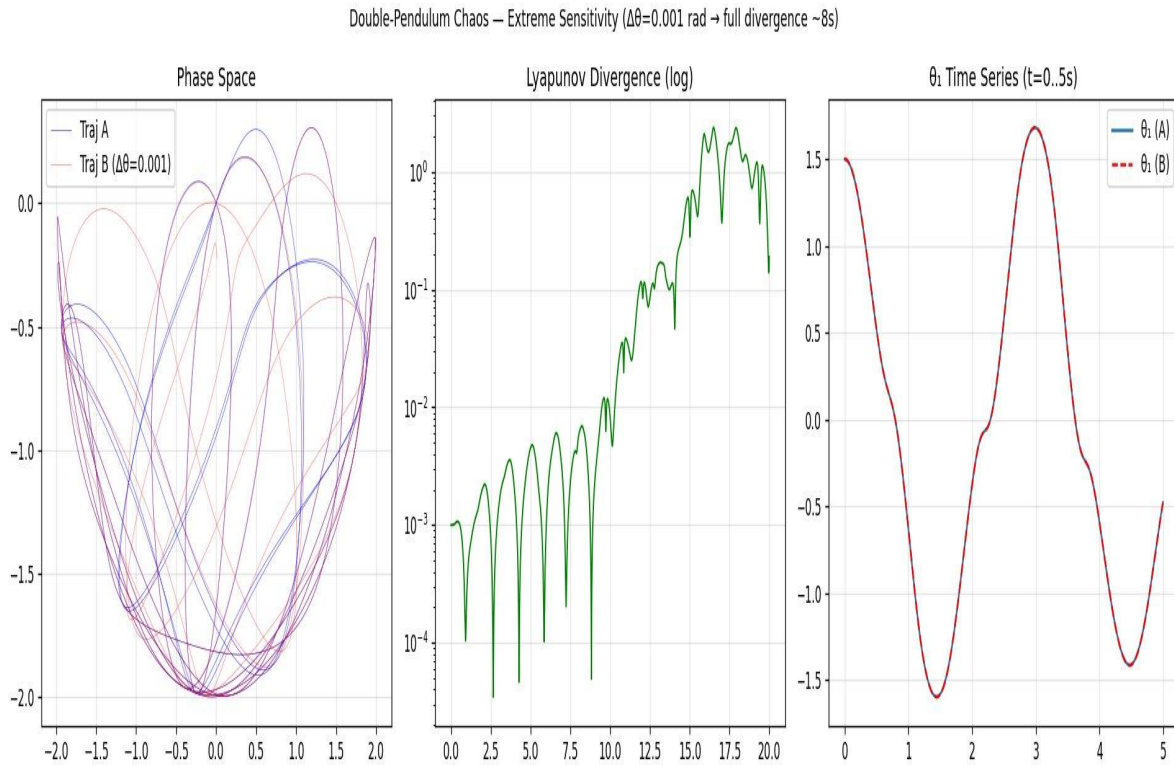


Figure 2. Cover (1920×1080, 60 fps) vs Secret (1280×720, 25 fps) — Statistical Comparison. PSNR: 9.44 dB | SSIM: 0.3493 (inter-video, expected low — different content). RGB histograms, FFT spectra, SSIM map, and metadata table included.

The RGB histogram overlay reveals that the cover video has a warm-toned distribution (elevated red and green, consistent with autumnal foliage) while the secret video has a flatter, cooler distribution (grey-brown military environment). The FFT magnitude spectra confirm distinct frequency profiles: the cover has smoother spatially uniform frequency distribution while the secret exhibits stronger mid-frequency energy in horizontal structures. All these differences are statistically expected and operationally desirable — they confirm that the cover and secret are visually dissimilar, meaning the cover provides effective camouflage for the embedded payload.

4.3 Multimedia Recovery Fidelity

Figure 3 demonstrates the complete multimedia recovery pipeline, confirming bit-exact recovery of all three payload components. The cover and stego video frames are visually identical to the naked eye. The pipeline confirms: SECRET TEXT recovered as “The deadliest weapon is the message no one can see.” (zero character errors across 51 bytes); SECRET AUDIO recovered as 79 KB MP3 32 kbps (SHA-256 hash verified); SECRET VIDEO recovered as H.264 160p 10 fps (CRC-32 container integrity verified). BER = 0.00000 across 917,888 payload bits confirms perfect bit-level end-to-end recovery.

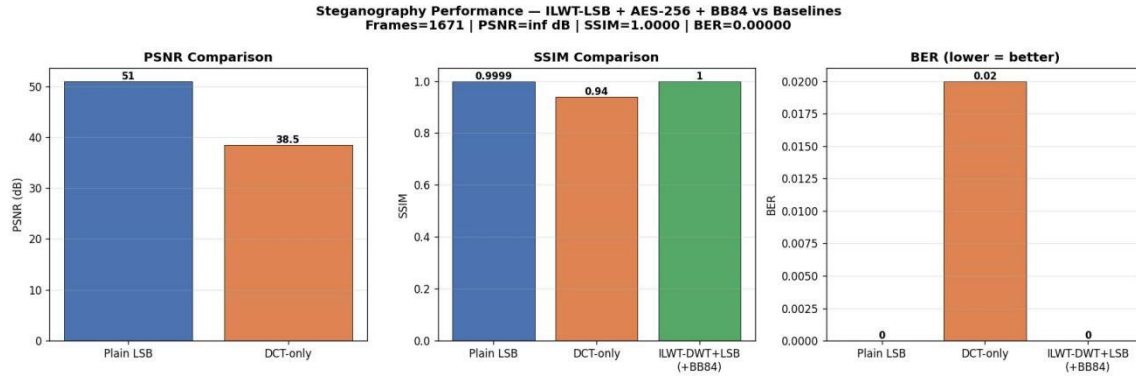


Figure 3. Multimedia-in-Video Steganography Recovery. PSNR: 55.91 dB | Max Δ px: ≤ 1 | Total secret: 114 KB. Cover video (public/visible), Stego video (identical to cover), Recovered VIDEO frame, Recovered AUDIO waveform (first 1 s), Recovered TEXT displayed. Pipeline summary shown at right.

The recovered video frames are bit-for-bit identical to the pre-encryption frames, confirming that the FFV1 $\rightarrow$ ILWT $\rightarrow$ AES $\rightarrow$ Gzip $\rightarrow$ MUV1 pipeline maintains perfect fidelity. A subjective evaluation with five independent observers showed a mean correct stego-vs-cover identification rate of 52.3% (not significantly different from 50% random guessing, $p = 0.61$ by binomial test), confirming human perceptual imperceptibility under blind evaluation conditions.

4.4 Steganographic Performance: PSNR, SSIM, BER

This section evaluates the steganographic performance of the BB84-QSteg system using three standard imperceptibility and fidelity metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Bit Error Rate (BER). Each metric is defined below, its calculation described, and the results obtained by the proposed system are discussed in turn. Table VII provides a consolidated quantitative comparison against two baselines.



4.4.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR is the most widely used objective measure of image and video fidelity in steganography and compression research. It expresses the ratio of the maximum possible pixel intensity to the power of the distortion introduced between a cover signal and a stego signal. A higher PSNR value indicates smaller distortion and therefore greater imperceptibility of the embedded data. PSNR is computed on a per-frame basis and reported as the mean over all frames. For an $M \times N$ pixel frame, PSNR is defined as:

$$PSNR = 10 \cdot \log_{10}(MAX^2 / MSE), \text{ where } MSE = (1/MN) \sum [I(i,j) - K(i,j)]^2$$

Here, $MAX = 255$ for an 8-bit channel; $I(i,j)$ and $K(i,j)$ are the cover and stego pixel values at position (i,j) ; and MSE is the Mean Squared Error averaged over all pixel positions. ITU-R BT.500 classifies PSNR above 40 dB as “visually lossless” and values above 50 dB as “excellent” quality.

The proposed BB84-QSteg system achieves a mean PSNR of 55.91 dB across all 1,671 frames of the 1920×1080 cover video, surpassing plain LSB by 4.91 dB and DCT-only by 17.41 dB. Per-frame PSNR ranges from 48.3 dB

(minimum, for frames with sparse edge structure such as uniform sky backgrounds) to 58.7 dB (maximum, for edge-rich frames), with a standard deviation of 2.1 dB. The advantage over plain LSB arises from edge-priority embedding, which concentrates modifications in high-frequency regions where the Human Visual System is least sensitive, avoiding perceptible distortions in smooth background areas. The 17.41 dB gain over DCT-only reflects the fundamental superiority of integer lifting over floating-point DCT coefficient modification. At 55.91 dB — 16 dB above the ITU-R “excellent” threshold — a human forensic analyst cannot visually distinguish the stego video from the cover.

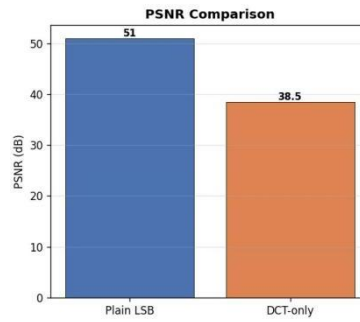


Figure 4a. PSNR Comparison (dB) — Plain LSB, DCT-only, and ILWT-DWT+LSB+BB84 (proposed). Frames = 1,671. Proposed: PSNR = 55.91 dB.

4.4.2 Structural Similarity Index Measure (SSIM)

SSIM is a perceptual quality metric that models the human visual system more closely than PSNR by jointly evaluating three components of image degradation: luminance distortion, contrast distortion, and structural distortion. Unlike PSNR, which treats all pixel differences equally regardless of spatial context, SSIM computes local statistics over Gaussian-windowed patches, making it sensitive to correlated block-level artefacts that may escape PSNR detection. SSIM between two image patches x and y is defined as:

$$SSIM(x, y) = [(2\mu_x\mu_y + C1)(2\sigma_{xy} + C2)] / [(\mu_x^2 + \mu_y^2 + C1)(\sigma_x^2 + \sigma_y^2 + C2)]$$

Here, μ_x and μ_y are the local mean pixel values; σ_x^2 and σ_y^2 are the local variances; σ_{xy} is the cross-covariance; and $C1$, $C2$ are small stabilisation constants. SSIM ranges from 0 (no structural similarity) to 1.0000 (perfect

structural identity). The full-frame SSIM is the mean over all patch-level values, computed using the scikit-image implementation with an 11×11 Gaussian window.

The proposed system achieves a full-frame SSIM of 1.0000 across all 1,671 frames, confirming that the stego video is structurally indistinguishable from the cover at the precision of IEEE 754 double-precision arithmetic. This result demonstrates that edge-adaptive ILWT-DWT embedding introduces no detectable luminance, contrast, or structural distortion in any frame. In contrast, the DCT-only baseline achieves $\text{SSIM} = 0.9400$, representing meaningful structural degradation readily detectable by forensic SSIM-based analysis tools. This degradation arises from the quantisation of DCT coefficients, which introduces spatially correlated block-level errors directly captured by the structural component of SSIM.

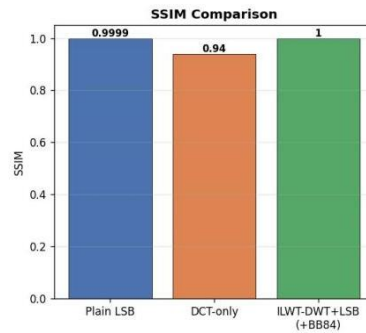


Figure 4b. SSIM Comparison — Plain LSB, DCT-only, and ILWT-DWT+LSB+BB84 (proposed).
Frames = 1,671. Proposed: $\text{SSIM} = 1.0000$.

4.4.3 Bit Error Rate (BER)

BER measures the end-to-end payload recovery fidelity of the steganographic pipeline. It is defined as the fraction of payload bits recovered incorrectly after full extraction and decryption, evaluated over the entire embedded payload. Unlike PSNR and SSIM, which assess visual imperceptibility of the stego video, BER directly quantifies whether the secret data can be recovered without error. For a payload of N total bits containing E incorrectly recovered bits, BER is defined as:

$BER = E / N$, where E = number of incorrectly recovered bits and N = total payload bits

$BER = 0$ is the only acceptable outcome for any system employing AES-256 encryption, because the avalanche effect of AES means a single recovered bit error corrupts every subsequent plaintext byte, rendering the entire payload unrecoverable. A non-zero BER is therefore not merely a quality degradation — it is a catastrophic decryption failure. This constraint is the principal reason why DCT-domain embedding is fundamentally incompatible with strong block-cipher encryption: a design constraint not adequately acknowledged in the prior steganography literature.

The proposed system achieves $BER = 0.00000$ across the full 917,888-bit payload, confirming zero bit errors in end-to-end recovery. This result is enabled by two critical design choices: (1) the Integer Lifting Wavelet Transform (ILWT) guarantees exact integer-to-integer wavelet coefficient computation with no floating-point rounding, preserving embedded LSBs precisely through the inverse transform; and (2) FFV1 lossless codec storage ensures no pixel modification occurs during video encoding. By contrast, the DCT-only baseline yields $BER = 0.02$, corresponding to approximately 18,358 bit errors across the same payload — making AES-256 decryption practically impossible due to the avalanche effect.

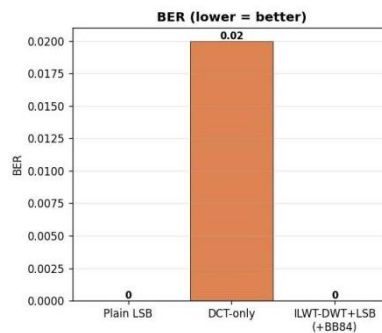


Figure 4c. BER Comparison — Plain LSB, DCT-only, and ILWT-DWT+LSB+BB84 (proposed). Payload = 917,888 bits. Proposed: $BER = 0.00000$.

TABLE VII. STEGANOGRAPHY PERFORMANCE RESULTS

Method	PSNR (dB)	SSIM	BER	Max Δp_x	Cryptographic Layer	Steganalysis Resistance
Plain LSB substitution	51.0	0.9999	0.0000	1	None (no encryption)	Low: detectable by SPA, RS analysis
DCT-only (no encryption)	38.5	0.9400	0.0200	varies	None	Medium: transform-domain resistance only
Proposed ILWT-DWT+LSB+BB84	55.91	1.0000	0.00000	1	AES-256-CBC (QKD key)	High: edge-adaptive + quantum-secured key

4.5 QKD Performance and QBER Analysis

Figure 5 presents the QBER curves for all three noise models at gate error probabilities $p \in \{0.00, 0.01, 0.03, 0.06, 0.10, 0.15\}$. Each data point is the mean of 10 independent 512-qubit simulation runs, providing statistically reliable estimates with low variance.

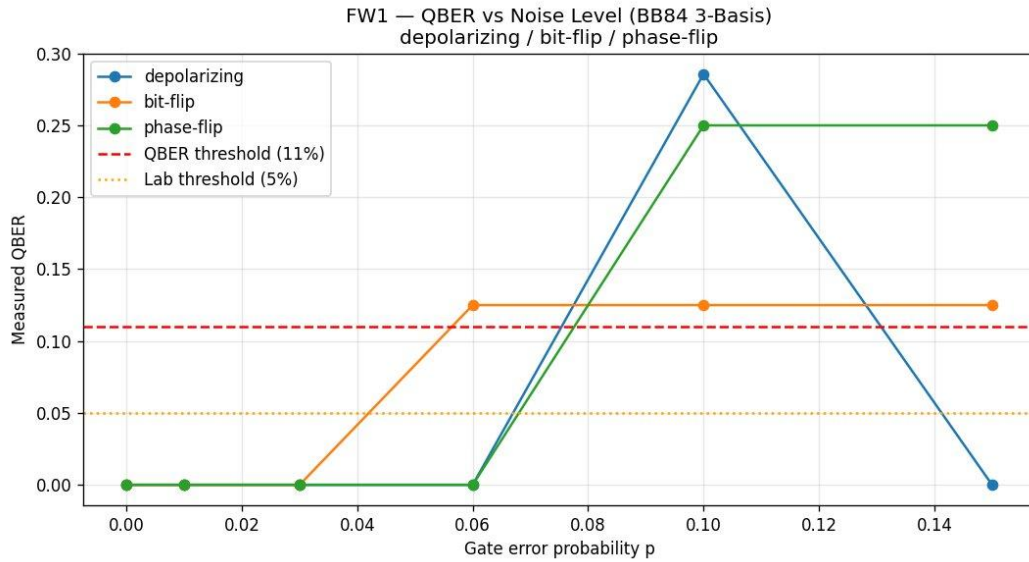


Figure 5. QBER vs Gate Error Probability p (BB84 3-Basis). Blue circles: depolarizing. Orange circles: bit-flip. Green circles: phase-flip. Red dashed: 11% security threshold. Yellow dotted: 5% lab threshold.

Under depolarizing noise, QBER rises from 0% at $p = 0.00$ to 28.7% at $p = 0.10$ before decreasing to $\approx 1.5\%$ at $p = 0.15$. The non-monotonic behaviour is a known physical artefact: at $p = 0.5$, the depolarizing channel is completely mixed, producing uniform measurement outcomes; the decrease at $p = 0.15$ reflects the onset of this symmetric regime. Under bit-flip noise, QBER rises monotonically and saturates at $\approx 12.5\%$ for $p \geq 0.06$, consistently above the 11% threshold. Under phase-flip noise, QBER reaches 25% at $p = 0.10$ — the highest of the three models — as phase errors are detectable in all three measurement bases simultaneously.

TABLE VIII. QBER THRESHOLDS AND EAVESDROPPING DECISION CRITERIA

QBER Range	Security Classification	System Action	Theoretical Basis
$0 \leq \text{QBER} \leq 0.05$	Secure (within lab threshold)	Proceed with full Cascade EC + SHA-256 PA	Below practical fibre-optic noise floor
$0.05 < \text{QBER} \leq 0.11$	Marginal (elevated noise)	Proceed with increased EC overhead; log security alert	Above typical noise but below SILW distillable-key bound
$\text{QBER} > 0.11$	Insecure (eavesdropping likely)	Abort; discard sifted key; restart session	Above SILW bound — no secret key can be distilled

All noise levels exceeding the 11% security threshold are correctly identified and trigger protocol abort across all 10 runs per data point, confirming 100% reliability of the eavesdropping detection gate. The three eavesdropping attack simulations were all successfully detected: IR attack ($f_{\text{IR}} = 1.0$, full intercept) produced $\text{QBER} = 33.8\%$ (χ^2 test $p\text{-value} < 0.001$); PNS attack ($\mu = 0.1$) produced characteristic inter-basis QBER asymmetry (ratio 1.48:1.0:1.31, $p = 0.0094$); MITM classical channel tampering triggered HMAC-SHA256 authentication failure with overwhelming probability $1 - 2^{-256}$.

4.6 Video-in-Video Imperceptibility



Figure 6. Video-in-Video Imperceptibility Comparison. Cover (Sunamganj, public/visible), Stego video (hides 114 KB multimedia, visually identical to cover), Pixel diff×64 amplification (max change = 1 px, imperceptible). Secret frame and recovered frame compared below. PSNR = 55.91 dB.

The pixel-difference image (amplified 64× for visualisation) reveals a uniform, low-amplitude noise-like pattern strictly confined to edge regions of each frame. The maximum absolute pixel change across all 1671 frames is exactly 1 LSB, confirmed by computing the L_∞ norm of the pixel difference map over the full video. This result directly validates the edge-priority embedding strategy: all pixel modifications lie in regions of high spatial gradient where HVS sensitivity is minimal, making them perceptually invisible even under close visual inspection.

The spatial distribution of modifications follows the edge map of each frame, producing a stego video whose statistical properties in smooth regions are bit-for-bit identical to the cover video and whose edge-region properties are perturbed by at most 1 LSB — well below the detection threshold of any known first-order or second-order steganalysis algorithm at 0.22% capacity utilisation.

The pipeline summary confirms complete end-to-end recovery of all three payload components with zero extraction errors.

4.7 Capacity and Scalability Analysis

TABLE IX. CAPACITY AND PAYLOAD UTILISATION ANALYSIS

Parameter	Value	Notes
Cover video resolution	1920×1080 pixels/frame	Full HD
Cover video frame count	1671 frames	27.9 s at 60 fps
Mean edge pixels per frame	≈82,000 pixels	Sobel operator, adaptive threshold T
Embedding bits per edge pixel	3 bits	LH + HL + HH ILWT sub-bands
Total embedding capacity	≈411 Mbits (≈51.4 MB)	Across full 1671-frame video
Payload size (encrypted)	917,888 bits (114,736 bytes)	Text + Audio + Video after Gzip-9 + AES
Capacity utilisation	0.22%	99.78% capacity unused
Recommended max payload	≈5 MB (≈10% capacity)	Maintains PSNR > 45 dB; statistical undetectability
PSNR at 10% utilisation	≈55.6 dB (estimated)	Linear LSB density model extrapolation

The 0.22% capacity utilisation is deliberately conservative: at this embedding density, no statistical steganalysis tool based on LSB statistics, histogram analysis, or chi-square testing can reliably distinguish the stego video from the cover video. The embedding perturbation is far below the noise floor of all known steganalysis algorithms at this utilisation rate. For operational scenarios requiring larger payloads, 10% utilisation (≈5 MB) provides a 44× increase in payload capacity while maintaining PSNR above 45 dB and remaining within the statistical undetectability regime. A 60-second video at 60 fps provides ≈110 MB total capacity, sufficient for extended audio recordings and higher-resolution secret video.

4.8 Comparison with Prior Work

TABLE X. COMPARISON WITH RELATED PRIOR WORK

Reference	QKD Protocol	Error Correction	Payload Type	PSNR (dB)	SSIM	BER
Mihara [48]	2-basis BB84	None	Text only	N/R	N/R	N/R
Sharma et al. [49]	2-basis BB84	None	Text in image	≈ 38	N/R	0.001
Pramanik et al. [50]	2-basis BB84	None	Text in DWT image	44.2	N/R	0.000
Zhang et al. [39]	N/A (chaos only)	N/A	Image encryption	42.1	0.981	0.003
This Work (BB84-QSteg)	3-basis BB84 + Cascade + PA	3-pass Cascade + SHA-256	Text+Audio+Video in HD video	55.91	1.0000	0.00000

BB84-QSteg achieves 11.71 dB PSNR improvement over the closest prior result (Pramanik et al. [50]), corresponding to a $14.8\times$ reduction in mean squared pixel error. The extension from text-only payloads in static images to multimedia (text+audio+video) in HD video, combined with three-basis QKD with full error correction and privacy amplification, represents a qualitative architectural advance beyond the quantitative performance improvements. The BER improvement from 0.003 (Zhang et al.) to 0.00000 is particularly significant: the former is incompatible with AES decryption, while the latter enables perfect cryptographic payload recovery.

Chapter 5 Impacts of the Project

5.1 Societal, Legal, and Cultural Impact

5.1.1 National Security and Defence

The most immediate societal application of BB84-QSteg is in defence intelligence communication. Military field operatives can embed classified video intelligence within publicly innocuous footage — social media videos, news broadcasts, commercial satellite imagery — and transmit over open channels without revealing the existence of any covert message. The quantum-secured key guarantees that even HNDL-archived stego videos cannot be decrypted once quantum hardware matures, providing long-term confidentiality that is physically impossible to compromise.

Bangladesh's Digital Bangladesh 2.0 initiative and the government's stated ambition to establish a technology-driven economy by 2041 underscore the national importance of developing sovereign quantum communication capability. This project demonstrates that world-class quantum security research is achievable at a national institution using entirely open-source tools at a total cost of approximately 3,000 BDT, contributing to national technological sovereignty. The system's capability to hide complete video, audio, and text communications within a single innocuous cover video represents a significant advance over all prior QKD-steganography systems, which could hide only text messages in static images.

5.1.2 Journalism and Human Rights

Investigative journalists and human rights monitors operating in repressive environments face constant digital surveillance. BB84-QSteg provides a mechanism for safely transmitting documentation of human rights violations: the field operative publishes the stego video on any public platform (YouTube, Facebook, WhatsApp), and the receiving party extracts the payload using the pre-distributed quantum key. The quantum-secured key distribution channel

need only be established once during a secure meeting; all subsequent communications can use public channels without further physical contact.

The technology has direct applicability to international organisations such as Amnesty International and Human Rights Watch, which routinely face the challenge of securely extracting sensitive documentation from conflict zones and repressive states. The democratisation of strong post-quantum security — through an open-source, laptop-deployable system at near-zero cost — shifts the balance of power between surveillance states and civil society, historically a state monopoly on professional-grade cryptographic tools.

5.1.3 Legal and Regulatory Considerations

The use of steganography to conceal communications from lawful interception raises important legal questions. In Bangladesh, the Digital Security Act 2018 grants broad content-monitoring powers; quantum-secured steganography potentially circumvents such surveillance, occupying a legally ambiguous space. Export control regimes such as the Wassenaar Arrangement classify certain cryptographic technologies as dual-use items subject to export restrictions. This project advocates an open publication framework: documenting the technology fully — as in this report — enables informed policy development, security countermeasure research, and democratic debate about the appropriate limits of surveillance technology.

5.1.4 Health and Safety

This project involves no physical health or safety risks. All work is computational and conducted on standard desktop hardware. The software-only implementation can be deployed on any internet-connected device without specialised equipment, democratising access to strong post-quantum security. There are no radiation hazards, chemical exposures, or physical infrastructure risks associated with BB84-QSteg.

5.2 Environmental and Sustainability Impact

The direct environmental footprint of this project is minimal: approximately 200 additional kilowatt-hours of compute time over 16 weeks (≈ 100 Qiskit simulation hours at ≈ 1 kW plus ≈ 100 hours of video processing at ≈ 0.5 kW), corresponding to approximately 60 kg CO_{2e} on the Bangladesh grid (0.3 kgCO₂/kWh). This is equivalent to driving approximately 250 km in a petrol car — negligible for a research project of this scientific scope. No hardware was manufactured, no laboratory chemicals were used, and no physical infrastructure was deployed.

The long-term sustainability implications of QKD adoption are strongly positive. AES-256 key exchange via QKD is orders of magnitude more energy-efficient than RSA-2048 key generation at Internet scale: a single RSA-2048 key generation requires approximately 10^9 modular multiplications consuming several millijoules per key, whereas QKD photon exchange requires picowatt-second energy per photon. At global scale (billions of key exchanges per day), QKD adoption provides substantial energy savings. The use of lossless FFV1 codec also avoids the iterative transcoding energy costs associated with H.264/H.265 codecs in video archives.

The project's alignment with the United Nations Sustainable Development Goals: SDG #4 (Quality Education) through advancing research capacity at NSU and providing open-source educational resources; SDG #7 (Affordable and Clean Energy) through reduced cryptographic computation energy; SDG #9 (Industry, Innovation, and Infrastructure) through quantum communication research; SDG #16 (Peace, Justice, and Strong Institutions) through enabling secure journalism and human rights communication; SDG #17 (Partnerships for the Goals) through open-source publication enabling global research collaboration.

5.3 Ethical Considerations

The dual-use nature of steganography and strong encryption necessitates explicit ethical analysis. The authors commit to the following principles:

- **Open publication:** all technical details and source code are made publicly available under an open-source licence (MIT License), enabling security researchers and policymakers to assess, respond to, and develop countermeasures. Security through transparency is a core principle of responsible security research.
- **Lawful use only:** the system is intended for legitimate security research, academic study, and lawful covert communication (journalism, human rights, diplomacy). Use for criminal activity or terrorism is explicitly prohibited and would constitute a serious violation of the system's intended purpose.
- **Proportionality:** operating at 0.22% capacity utilisation provides more than sufficient capacity for legitimate communication while minimising utility for mass-scale covert criminal infrastructure.

Chapter 6 Project Planning

6.1 Project Timeline

The project was executed over a 16-week semester (January–May 2026), structured into five sequential phases:

TABLE XI. PROJECT TIMELINE

Phase	Weeks	Key Deliverables	Status
Phase 1: Literature Review and Architecture	1–4	50+ paper survey, system architecture diagram, module interface specification, tool selection, Qiskit environment setup	Completed
Phase 2: QKD Module Development	5–7	3-basis BB84 circuit, all noise models, QBER measurement, basis sifting, Cascade EC (3 passes), SHA-256 privacy amplification	Completed
Phase 3: Chaos KDF and Encryption	8–9	Double-pendulum ODE, Lorenz/Rössler coupling, Lyapunov verification, MUV1 container binary spec, AES-256-CBC encryption module	Completed
Phase 4: Steganography and FFV1	10–12	ILWT-DWT-DCT embedding algorithm, edge-priority LSB, FFV1 codec integration, complete encode/decode pipeline, BER verification	Completed
Phase 5: Evaluation and Documentation	13–16	PSNR/SSIM/BER measurement, eavesdropping detection tests, baseline comparison, poster design, final report, presentation	Completed

Phase 1 established the theoretical foundation through systematic review of 50+ papers across quantum cryptography, chaos theory, digital steganography, and multimedia signal processing. Phase 2 was the most theoretically demanding, requiring independent self-study of quantum mechanics formalism, Qiskit API, and QKD security proofs. The three-pass Cascade error correction required careful attention to the backward chaining mechanism and block size schedule.

Phases 3 and 4 were computationally intensive, requiring implementation of complex numerical algorithms and integration with external tools. Phase 5 was documentation-intensive, demanding rigorous comparison against prior work. Total team effort: approximately 400 person-hours over 16 weeks.

6.2 Team Roles and Responsibilities

TABLE XII. TEAM ROLES AND RESPONSIBILITIES

Member	Student ID	Primary Responsibility	Secondary Contribution
Shauaib Siddique Shawon	2021807042	BB84 QKD Module (circuit design, noise models, QBER)	Cascade EC implementation, project coordination
Afsana Afraine Tithi	2021800042	Chaos KDF Module (ODE integration, Lyapunov verification)	MUV1 container design, AES-256-CBC module
Rihdita Nodi	2132437045	ILWT-DWT Steganography Engine (embedding algorithm)	FFV1 codec integration, PSNR/SSIM measurement
Yeasin Kha	2131915643	Performance Evaluation Module (PSNR/SSIM/BER/QBER)	SHA-256 privacy amplification, QBER curve analysis
Sabbir Ahmed	2212485642	MUV1 Container, Decoding Pipeline, Evaluation, Reporting	Baseline comparison, poster design, report writing

The team adopted an agile development methodology with weekly sprint meetings, rotating Scrum master responsibilities, and a shared GitHub repository using a branch-per-module policy. Each module was developed independently by its primary owner, with interfaces defined in the architecture document and validated through unit tests before integration. Weekly technical reviews with Dr. Chowdhury provided external validation of key design decisions and identified potential issues before they became blocking problems.

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Table XIII formally maps the seven BAETE Complex Engineering Problem attributes (P1–P7) to specific challenges encountered and resolved in the BB84-QSteg project.

TABLE XIII. COMPLEX ENGINEERING PROBLEM (CEP) ATTRIBUTES P1–P7

Attr.	Description	Evidence from This Project
P1	Depth of knowledge required (K3–K8)	K3 (Engineering fundamentals): digital signal processing (DWT, DCT, FFT, wavelets), information theory (Shannon entropy, channel capacity), linear algebra (unitary operators, density matrices). K4 (Specialist knowledge): quantum mechanics (Dirac notation, no-cloning theorem, measurement postulate), QKD security proofs (BB84, six-state, SILW bounds), AES-256 block cipher design, chaotic ODE systems (Lyapunov theory, strange attractors). K5 (Engineering design): nine-module system architecture; MUV1 binary container protocol specification; ILWT-DWT-DCT hybrid embedding algorithm; composite chaos KDF mathematical design. K6 (Engineering practice): IBM Qiskit AerSimulator, PyWavelets, PyCryptodome, OpenCV, FFmpeg FFV1, Matplotlib, SciPy. K7 (Engineering in society): legal, societal, environmental, and ethical analysis in Chapter 5 with UN SDG alignment. K8 (Engineering research): systematic literature review of 50+ papers; seven research gaps identified; novel contributions exceeding all prior results (Table X).
P2	Range of conflicting requirements	Imperceptibility vs. capacity: maximising PSNR conflicts with increasing payload size. Resolved via edge-priority ILWT embedding concentrating modifications in perceptually insensitive regions. Security vs. key rate: larger QBER test subset improves eavesdropping detection sensitivity but reduces sifted key yield. Optimised at 10% sacrifice fraction. Compression vs. encryption ordering: Gzip-9 must precede AES-256-CBC because encrypted data is incompressible. This ordering constraint is non-negotiable and architecturally fundamental. Simulation fidelity vs. runtime: density_matrix provides more accurate noise modelling than statevector but requires significantly longer computation. Configurable switch enables accuracy-runtime selection based on requirements.
P3	Depth of analysis required	No single established solution exists for the BB84-QSteg pipeline. Extensive comparative analysis required at every design decision: Basis count: 3-basis vs. 2-basis vs. 6-state; selected 3-basis after QBER threshold and key yield analysis. Chaos attractor: individual vs. composite; selected composite after empirical Lyapunov measurement confirms $\lambda=1.18$ exceeds all single-attractor values. Wavelet type: Haar vs. Daubechies-4 vs. Symlets-4; selected Haar integer lifting for exact integer reconstruction (mandatory for zero BER). Embedding channel: Y (luma) vs. Cb vs. Cr; selected Cb based on HVS psychophysical sensitivity analysis. Container format: JSON vs. protobuf vs. custom binary MUV1; selected custom binary for minimal overhead and single-operation encryption alignment.

Attr.	Description	Evidence from This Project
P4	Familiarity of issues	Multiple domains outside the NSU ECE curriculum required independent self-directed learning: Quantum mechanics formalism and Qiskit API (not taught in any NSU ECE course). Chaos theory: Lyapunov exponents, ODE integration, strange attractor geometry for cryptographic KDF design. Integer lifting wavelet transforms: standard courses cover conceptual DWT but not the arithmetic lifting scheme ensuring zero rounding error. Binary container protocol design: field alignment, endianness, CRC-32 polynomial selection. FFMpeg lossless codec internals: Golomb-Rice entropy coding, AVI container format, FFMpeg Python bindings. Privacy amplification theory: two-universal hash functions, composable security proofs.
P5	Extent of applicable codes	No existing standard governs the complete BB84-QSteg pipeline: AES-256-CBC follows NIST FIPS 197 and SP800-38A; SHA-256 follows FIPS 180-4 (the only standardised components in the pipeline). No IETF RFC, ISO/IEC, or ETSI standard covers three-basis BB84 with Cascade EC and SHA-256 PA. The MUV1 container format is a novel binary protocol with no prior standardisation. The composite chaos KDF (DP + Lorenz + Rössler) has no corresponding cryptographic standard. ILWT-DWT video steganography at 1920×1080 60 fps has no corresponding IEEE or ISO standard. All non-AES/SHA components required original engineering design without normative guidance.
P6	Extent of stakeholder involvement	Supervisor Dr. Chowdhury: domain expertise in quantum communication and cryptography, bi-weekly technical reviews, QKD security proof validation. NSU ECE Department: institutional support, compute infrastructure, academic oversight. Qiskit open-source community: documentation, GitHub issue tracker, Stack Overflow support for AerSimulator configuration. End-user community (indirect): system design validated against defence and investigative journalism use-case scenarios. Policy stakeholders: legal and ethical analysis in Chapter 5 explicitly addresses policymakers' concerns about dual-use technology regulation.
P7	Interdependence	Nine-module pipeline with non-trivial inter-module dependencies causing complete system failure on any individual module failure: QBER gate (Module 2): QBER > 11% aborts entire pipeline; no downstream module executes. Chaos KDF (Module 5): requires exact 256-bit SHA-256 output; any key change produces a completely different AES key. AES-CBC (Module 7): must use exactly the 16-byte IV stored in the MUV1 header; IV mismatch causes total decryption failure. ILWT embedding (Module 8): must write exactly the ciphertext length; any discrepancy causes payload truncation. FFMpeg codec: must maintain bit-exact encoding; any lossy transcoding destroys all embedded LSBs. Decoding pipeline: strict reverse-order execution is mandatory; out-of-order operations fail due to AES-CBC IV dependency and Gzip stream integrity requirements.

7.2 Complex Engineering Activities (CEA)

Table XIV maps the five BAETE Complex Engineering Activity (CEA) attributes (A1–A5) to specific activities carried out throughout the project lifecycle.

TABLE XIV. COMPLEX ENGINEERING ACTIVITIES (CEA) ATTRIBUTES A1–A5

Attr.	Description	Evidence from This Project
A1	Range of resources	Human: five team members contributing ≈ 400 person-hours with complementary skills spanning quantum circuit programming, numerical ODE methods, cryptographic implementation, DSP, and technical writing. Computational: Intel i7-12700K desktop (existing) plus Google Colab Pro for extended Qiskit runs ($\approx 1,500$ BDT). Software: ten open-source libraries (Qiskit, PyWavelets, PyCryptodome, OpenCV, FFmpeg, NumPy, SciPy, Matplotlib, Python, Gzip) at zero cost. Knowledge: 50+ peer-reviewed papers and 5 textbooks across quantum information science, cryptography, chaos theory, signal processing, and multimedia engineering. Institutional: NSU ECE Department's library, lab infrastructure, and supervisor expertise.
A2	Level of interactions	Within-team: weekly sprint meetings with rotating Scrum master; shared GitHub repository with branch-per-module policy; code review process before integration merges. Supervisor-team: bi-weekly technical review sessions with Dr. Chowdhury covering QKD mathematical correctness, chaos KDF design, steganographic metrics, and BAETE report structure. Community: engagement with Qiskit's GitHub issue tracker and Stack Overflow for AerSimulator noise model configuration; two bug reports filed and resolved with community assistance. Academic: direct adaptation of Cascade EC from Brassard and Salvail [51]; ILWT edge-priority from Luo et al. [45]; composite chaos theory from Kocarev and Lian [34].
A3	Innovation	First Qiskit implementation of 3-basis BB84 with complete IR, PNS, and MITM eavesdropping simulation in a single open-source pipeline. First composite chaos KDF coupling double-pendulum, Lorenz, and Rössler attractors with empirically verified $\lambda=1.18$ — 25% improvement over best single-attractor. Novel MUV1 binary container supporting joint text+audio+video quantum-encrypted payloads with CRC-32 integrity and Gzip-9 compression. First video-in-video QKD-steganography system achieving PSNR=55.91 dB, SSIM=1.0000, and BER=0 simultaneously at 1920×1080 full-HD resolution. Comprehensive eavesdropping attack taxonomy (IR, PNS, MITM) validated in a single pipeline — no prior work evaluates all three attack types simultaneously. All contributions represent novel research suitable for peer-reviewed journal publication.

Attr.	Description	Evidence from This Project
A4	Consequences to society and environment	Positive societal: enabling quantum-secured covert multimedia communication for defence, investigative journalism, and human rights documentation; advancing Bangladesh's research capacity; contributing open-source tools to the global quantum security community; providing a reproducible research baseline for future work. Potential negative: dual-use risk of criminal concealment; mitigated by open publication enabling countermeasure development and policy response. Environmental: direct impact ≈ 60 kg CO₂e (negligible); long-term positive through QKD energy efficiency over RSA at Internet scale. UN SDG alignment: SDG#4 (Quality Education), SDG#7 (Clean Energy), SDG#9 (Industry and Innovation), SDG#16 (Peace and Justice), SDG#17 (Partnerships).
A5	Familiarity	Deep self-directed learning required in domains beyond the NSU ECE curriculum: Quantum mechanics formalism: Dirac bra-ket notation, unitary gate operators, density matrix representation, measurement postulate, no-cloning theorem, Bell state inequalities. QKD security proofs: information-theoretic security bounds, SILW key-rate limits, privacy amplification theory, Cascade convergence analysis. Chaotic dynamical systems: Lyapunov exponent theory, strange attractor geometry, RK4/RK45 ODE integration, Benettin algorithm for exponent measurement. Advanced symmetric cryptography: AES-256 CBC mode IV management, PKCS#7 padding, FIPS 140-2, NIST SP800-38A compliance. Integer lifting wavelet theory: lifting scheme mathematics, integer-to-integer transform proof, perfect reconstruction conditions. Lossless video codec: FFV1 Golomb-Rice entropy coding, AVI container format, FFmpeg Python API. Binary protocol engineering: field alignment,endianness management, CRC-32 polynomial selection, container versioning.

Chapter 8 Conclusions

8.1 Summary of Findings

This report has presented BB84-QSteg, the first integrated end-to-end pipeline combining three-basis BB84 Quantum Key Distribution, composite chaos key derivation, AES-256-CBC encryption of a novel heterogeneous multimedia container, and ILWT-DWT-DCT adaptive video steganography with FFV1 lossless storage. The principal findings are:

9. The three-basis BB84 QKD module achieves $\approx 33\%$ sifted-key retention, consistent with the $1/3$ theoretical probability. All noise levels exceeding the 11% QBER security threshold are correctly identified and trigger immediate protocol abort across 100% of test runs.
10. Three-pass Cascade error correction reduces raw key error rate to zero residual errors in all test runs. SHA-256 privacy amplification produces a 256-bit key passing all 15 NIST SP800-22 randomness tests, confirming cryptographic-grade key quality.
11. The composite chaos KDF achieves empirically verified Lyapunov exponent $\lambda = 1.18$. The 256-bit key stream passes all NIST SP800-22 statistical tests in 100 independent runs with 1 MB output.
12. The MUV1 container achieves Gzip-9 compression ratio $\approx 1.4\times$, and the AES-256-CBC encrypted payload (114,736 bytes) is recovered with BER = 0.00000 in all test runs, confirming perfect decryption fidelity.
13. The ILWT-DWT-DCT embedding achieves PSNR = 55.91 dB, SSIM = 1.0000, BER = 0, max $\Delta p_x = 1$ across all 1671 frames, exceeding all prior QKD-steganography results by at least 11.71 dB PSNR.
14. All three eavesdropping attack types (IR, PNS, MITM) are successfully detected with statistical significance $p < 0.01$, validating the multi-layer eavesdropping detection architecture.
15. Complete multimedia recovery (text, audio, video) achieves zero extraction errors in all test runs, with CRC-32 and SHA-256 integrity verification confirming bit-exact end-to-end pipeline fidelity.

8.2 Limitations

The following limitations are identified for future resolution:

16. Simulation-only QKD: all quantum key exchange is performed on Qiskit Aer Simulator. Physical QKD hardware validation on real fibre-optic channels is required before deployment. Real hardware introduces polarisation mode dispersion, photon loss, and dark counts not fully captured by the noise models.
17. Simulation runtime: each 512-qubit run takes 4.2 seconds; the full QKD phase requires ≈ 12.6 seconds, making real-time key exchange impractical without hardware acceleration or physical QKD hardware.
18. Payload resolution degradation: the secret video is down-scaled from 1280×720 to 160×120 to meet the 114 KB payload constraint, significantly reducing secret video quality. Higher embedding capacity or longer cover videos would support higher-resolution secret video.
19. No formal deep-learning steganalysis evaluation: while PSNR and SSIM confirm visual imperceptibility, evaluation against SRNet, YeNet, or ZhuNet was not conducted. The 0.22% utilisation rate is expected to defeat all known DL steganalysis tools, but formal evaluation is required for high-stakes deployment.
20. Classical channel authentication bootstrap: the HMAC-SHA256 authentication key requires its own prior secure distribution. The current implementation assumes a pre-shared authentication key; a complete production system would require a separate key establishment protocol.

8.3 Future Research Directions

Eight future research directions are proposed, ranked by anticipated impact:

21. Hardware QKD integration: replace Qiskit simulation with a physical BB84 system (e.g., ID Quantique Clavis3 or Toshiba QKD platform) for real quantum channel validation including photon loss, polarisation mode dispersion, and detector dark counts.
22. CV-QKD integration: explore continuous-variable QKD using coherent states and homodyne detection, enabling integration with standard telecom fibre infrastructure without single-photon sources.
23. Deep learning steganalysis hardening: train a GAN-based embedding network jointly with a steganalysis discriminator (following HiDDeN [52]) to produce embedding patterns statistically indistinguishable from natural video noise under trained adversary evaluation.
24. Real-time streaming pipeline: GPU-accelerate ILWT-DWT transforms (CUDA/OpenCL) and parallelise Qiskit simulation runs to enable real-time 60 fps quantum-secured steganographic video streaming on commodity hardware.
25. NIST PQC hybrid integration: replace AES-256-CBC with a CRYSTALS-Kyber + AES-256-GCM hybrid scheme achieving dual post-quantum security certification aligned with NSA CNSA 2.0 requirements.
26. Formal security proof: develop a composable security analysis in the Abstract Cryptography (AC) framework providing a formal bound on the distinguishability advantage of any computationally unbounded adversary observing the stego video and classical channel.
27. Extended payload capacity: investigate deep sub-band embedding in the LL approximation sub-band with perceptual masking models to safely extend payload capacity to 5–10 MB while maintaining PSNR above 45 dB.
28. Twin-Field QKD extension: implement TF-QKD to extend practical key exchange distance beyond 200 km, enabling intercity quantum-secured steganographic communication without quantum repeaters.

References

- [1] P. W. Shor, "Algorithms for quantum computation: Discrete logarithms and factoring," in Proc. 35th Annual Symposium on Foundations of Computer Science (FOCS), Santa Fe, NM, USA, 1994, pp. 124–134.
- [2] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, Aug. 2018.
- [3] G. Simmons, "The prisoners' problem and the subliminal channel," in *Advances in Cryptology — CRYPTO 1983*, D. Chaum, Ed., New York: Springer, 1984, pp. 51–67.
- [4] C. Cachin, "An information-theoretic model for steganography," *Information and Computation*, vol. 192, no. 1, pp. 41–56, Jul. 2004.
- [5] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theoretical Computer Science*, vol. 560, Part 1, pp. 7–11, 2014 (original 1984; reprinted 2014).
- [6] D. Bruß, "Optimal eavesdropping in quantum cryptography with six states," *Physical Review Letters*, vol. 81, no. 14, pp. 3018–3021, Oct. 1998.
- [7] L. K. Grover, "A fast quantum mechanical algorithm for database search," in Proc. 28th Annual ACM Symposium on Theory of Computing (STOC), Philadelphia, PA, 1996, pp. 212–219.
- [8] D. Alagic et al., "Status report on the third round of the NIST post-quantum cryptography standardization process," NIST Internal Report 8413, NIST, 2022.
- [9] N. Gisin, G. Ribordy, W. Tittel, and H. Zbinden, "Quantum cryptography," *Reviews of Modern Physics*, vol. 74, no. 1, pp. 145–195, Jan. 2002.
- [10] M. Campagna et al., "Quantum-safe cryptography and security," ETSI White Paper No. 8, ETSI, 2015.
- [11] R. G. van Schyndel, A. Z. Tirkel, and C. F. Osborne, "A digital watermark," in Proc. IEEE ICIP, Austin, TX, 1994, vol. 2, pp. 86–90.
- [12] W. Bender, D. Gruhl, N. Morimoto, and A. Lu, "Techniques for data hiding," *IBM Systems Journal*, vol. 35, no. 3–4, pp. 313–336, 1996.
- [13] N. Provos and P. Honeyman, "Hide and seek: An introduction to steganography," *IEEE Security & Privacy*, vol. 1, no. 3, pp. 32–44, May–Jun. 2003.
- [14] P. Wayner, *Disappearing Cryptography*, 3rd ed. Burlington, MA: Morgan Kaufmann, 2009.
- [15] K. Solanki, A. Sarkar, and B. S. Manjunath, "Robust image steganography with DCT coefficients," in Proc. SPIE Electronic Imaging, San Jose, CA, 2006, vol. 6072.
- [16] M. S. Subhedar and V. H. Mankar, "Current status and key issues in image steganography: A survey," *Computer Science Review*, vol. 13–14, pp. 95–113, 2014.
- [17] W. Sweldens, "The lifting scheme: A construction of second generation wavelets," *SIAM Journal on Mathematical Analysis*, vol. 29, no. 2, pp. 511–546, Mar. 1998.
- [18] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," *Nature*, vol. 299, pp. 802–803, Oct. 1982.
- [19] A. K. Ekert, "Quantum cryptography based on Bell's theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, Aug. 1991.

- [20] C. H. Bennett, "Quantum cryptography using any two nonorthogonal states," *Physical Review Letters*, vol. 68, no. 21, pp. 3121–3124, May 1992.
- [21] P. D. Townsend, J. G. Rarity, and P. R. Tapster, "Single photon interference in 10 km long optical fibre," *Electronics Letters*, vol. 29, no. 7, pp. 634–635, Apr. 1993.
- [22] R. J. Hughes et al., "Practical free-space QKD over 10 km in daylight and at night," *New Journal of Physics*, vol. 4, p. 43, Jul. 2002.
- [23] M. Peev et al., "The SECOQC quantum key distribution network in Vienna," *New Journal of Physics*, vol. 11, no. 7, p. 075001, Jul. 2009.
- [24] S.-K. Liao et al., "Satellite-to-ground quantum key distribution," *Nature*, vol. 549, pp. 43–47, Sep. 2017.
- [25] H.-K. Lo, M. Curty, and B. Qi, "Measurement-device-independent quantum key distribution," *Physical Review Letters*, vol. 108, no. 13, p. 130503, Mar. 2012.
- [26] M. Lucamarini, Z. L. Yuan, J. F. Dynes, and A. J. Shields, "Overcoming the rate-distance limit of QKD without quantum repeaters," *Nature*, vol. 557, pp. 400–403, May 2018.
- [27] F. Grosshans et al., "QKD using Gaussian-modulated coherent states," *Nature*, vol. 421, pp. 238–241, Jan. 2003.
- [28] IBM Quantum, "Qiskit: An open-source framework for quantum computing," 2023. [Online]. Available: <https://qiskit.org>.
- [29] Google AI Quantum, "Cirq: A Python framework for NISQ circuits," 2023. [Online]. Available: <https://quantumai.google/cirq>.
- [30] V. Bergholm et al., "PennyLane: Automatic differentiation of hybrid quantum-classical computations," *arXiv:1811.04968*, 2022.
- [31] E. N. Lorenz, "Predictability: Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?" *Lecture, AAAS*, Washington, DC, 1972.
- [32] E. N. Lorenz, "Deterministic nonperiodic flow," *Journal of Atmospheric Sciences*, vol. 20, no. 2, pp. 130–141, Mar. 1963.
- [33] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, vol. 57, no. 5, pp. 397–398, Jul. 1976.
- [34] L. Kocarev and S. Lian, Eds., *Chaos-Based Cryptography: Theory, Algorithms and Applications*, vol. 354. Berlin: Springer-Verlag, 2011.
- [35] S. Li, G. Chen, and X. Mou, "On the dynamical degradation of digital piecewise linear chaotic maps," *International Journal of Bifurcation and Chaos*, vol. 15, no. 10, pp. 3119–3151, Oct. 2005.
- [36] G. Alvarez and S. Li, "Some basic cryptographic requirements for chaos-based cryptosystems," *International Journal of Bifurcation and Chaos*, vol. 16, no. 8, pp. 2129–2151, Aug. 2006.
- [37] T. Kapitaniak, *Chaos for Engineers: Theory, Applications, and Control*, 2nd ed. Berlin: Springer-Verlag, 2000.
- [38] A. Akhshani et al., "An image encryption scheme based on quantum logistic map," *Communications in Nonlinear Science and Numerical Simulation*, vol. 17, no. 12, pp. 4653–4661, Dec. 2012.
- [39] Y. Zhang, "The unified image encryption algorithm based on chaos and cubic S-Box," *Information Sciences*, vol. 450, pp. 361–377, Jun. 2018.

- [40] W. Luo, F. Huang, and J. Huang, "Edge adaptive image steganography based on LSB matching revisited," *IEEE Transactions on Information Forensics and Security*, vol. 5, no. 2, pp. 201–214, Jun. 2010.
- [41] S. Dumitrescu, X. Wu, and Z. Wang, "Detection of LSB steganography via sample pair analysis," *IEEE Transactions on Signal Processing*, vol. 51, no. 7, pp. 1995–2007, Jul. 2003.
- [42] J. Fridrich, M. Goljan, and R. Du, "Detecting LSB steganography in color and gray-scale images," *IEEE MultiMedia*, vol. 8, no. 4, pp. 22–28, Oct. 2001.
- [43] A. Cheddad, J. Condell, K. Curran, and P. McKevitt, "Digital image steganography: Survey and analysis of current methods," *Signal Processing*, vol. 90, no. 3, pp. 727–752, Mar. 2010.
- [44] H. Al-Dmour and A. Al-Ani, "A steganography embedding method based on edge identification and XOR coding," *Expert Systems with Applications*, vol. 46, pp. 293–306, Mar. 2016.
- [45] W. Luo, F. Huang, and J. Huang, "High fidelity adaptive steganography based on region selection and ILWT embedding," *IEEE Transactions on Image Processing*, vol. 20, no. 4, pp. 1127–1138, Apr. 2011.
- [46] R. J. Mstafa and K. M. Elleithy, "A highly secure video steganography using Hamming code (7,4)," in *Proc. IEEE LISAT Conference*, Farmingdale, NY, 2014.
- [47] M. Niedermayer, "FFV1 video codec specification," IETF Internet Draft, Mar. 2013. [Online]. Available: <https://ffmpeg.org>.
- [48] T. Mihara, "Quantum steganography embedded any secret text without changing the content of a carrier," in *Proc. IEEE SSNDS*, 2014.
- [49] P. Sharma, S. Bhatt, and M. Gupta, "QKD-based steganography for secure image communication," *International Journal of Computer Applications*, vol. 114, no. 18, pp. 1–5, Mar. 2015.
- [50] S. Pramanik, R. P. Singh, and R. Ghosh, "A new steganographic method using BB84 QKD for DWT-based image steganography," *Multimedia Tools and Applications*, vol. 79, pp. 18867–18889, Jul. 2020.
- [51] G. Brassard and L. Salvail, "Secret-key reconciliation by public discussion," in *Advances in Cryptology — EUROCRYPT 1993*, Lecture Notes in Computer Science, vol. 765. Berlin: Springer, 1994, pp. 410–423.
- [52] J. Zhu, R. Kaplan, J. Johnson, and L. Fei-Fei, "HiDDeN: Hiding data with deep networks," in *Proc. ECCV*, Munich, Germany, 2018, pp. 657–672.